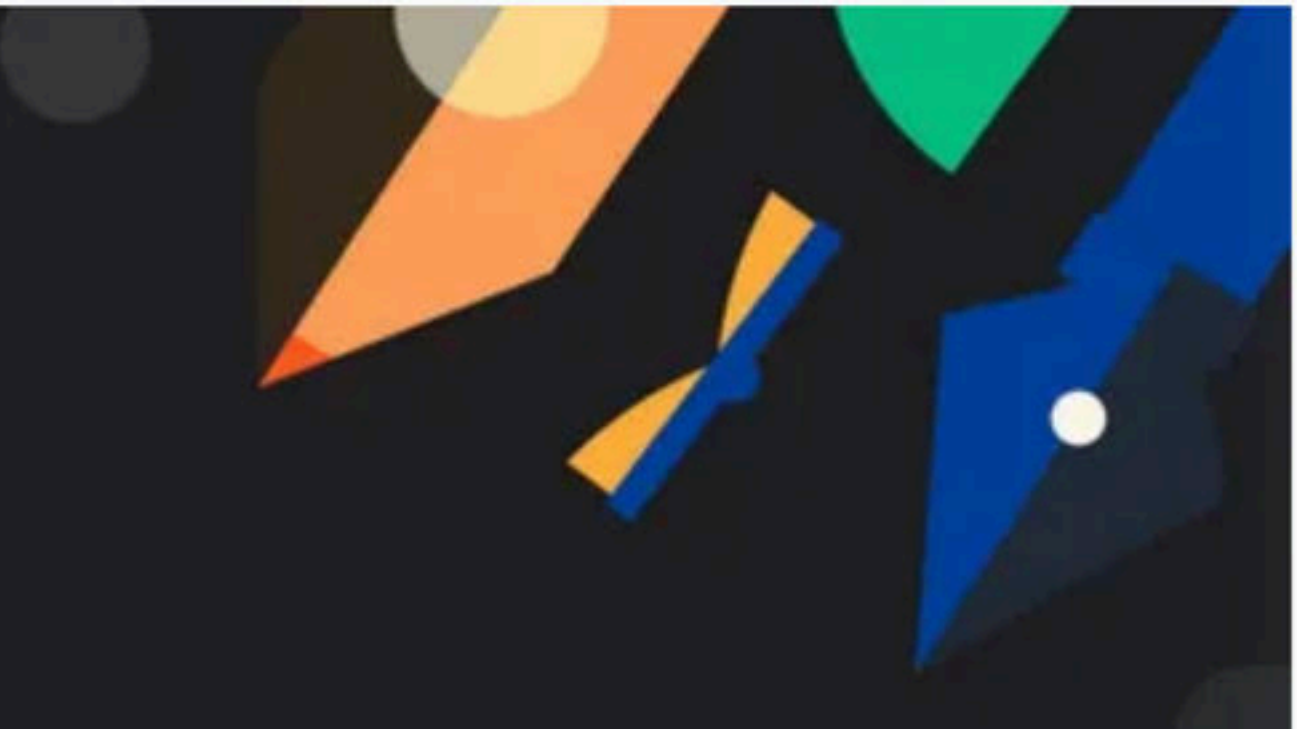




Doubt Class With Lakshay

Special class

Generic tree

BST

BT

Binary Search Trees Class - 1

Special class

BST



sorted

⇒ BST

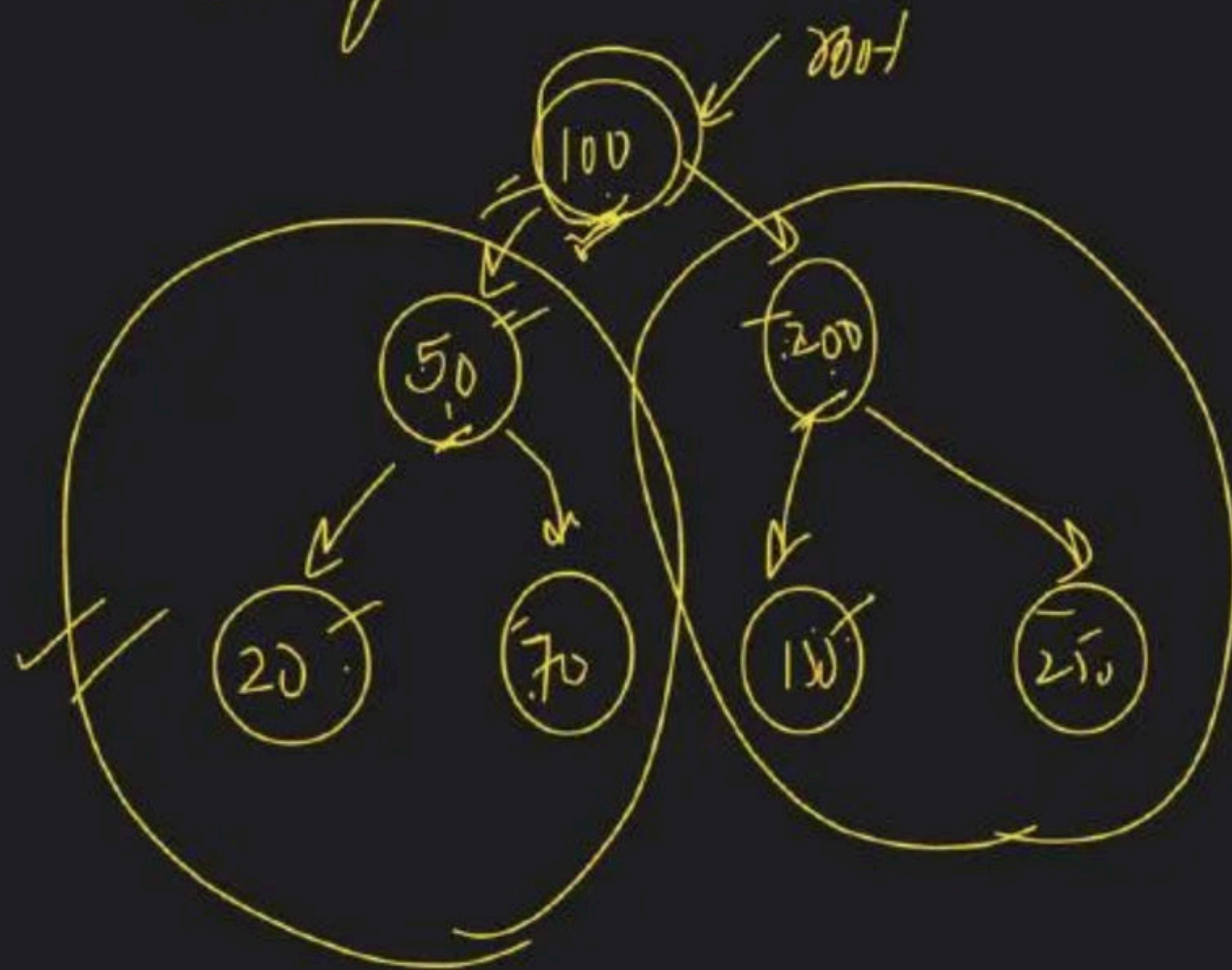
5 min

Break

↳ uploading tree

class rec...

BST $\rightarrow$ Binary Search Tree



for every
node

↓
property

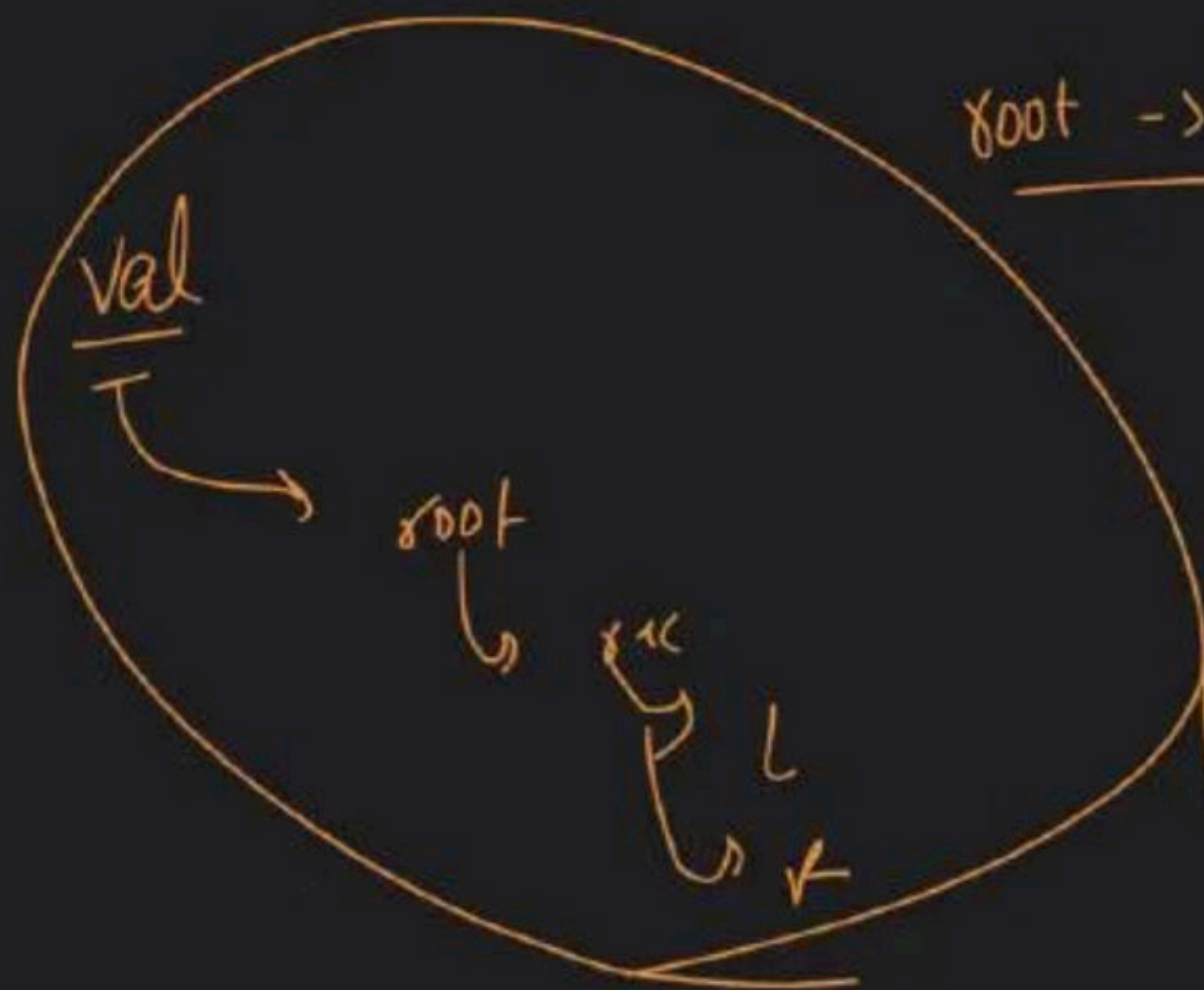
↓
left val < root val
child.

right
child val > root
val

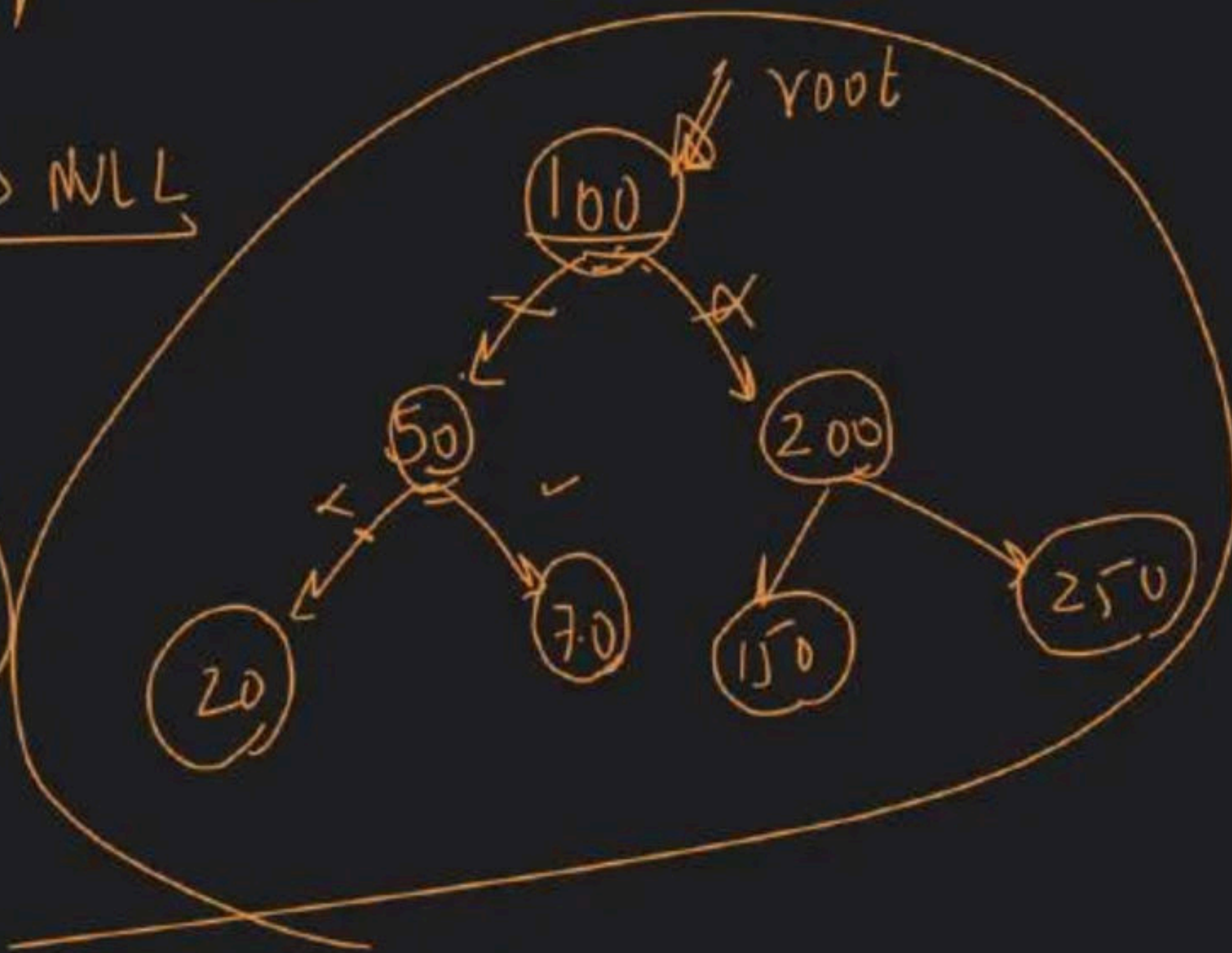
BST

if $\rightarrow$ 100, 50, 200, 70, 20, 250, 150, 1

rukun

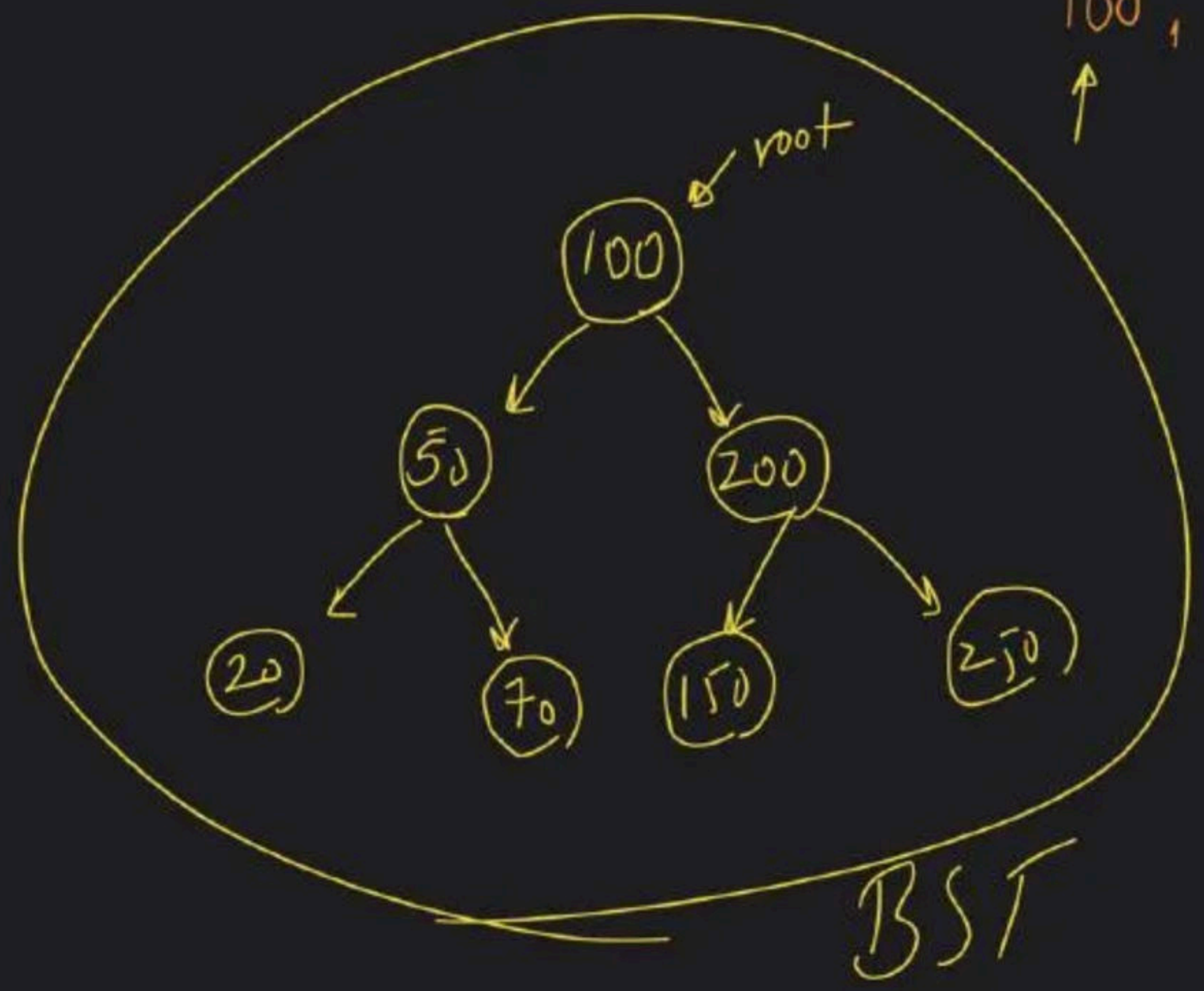


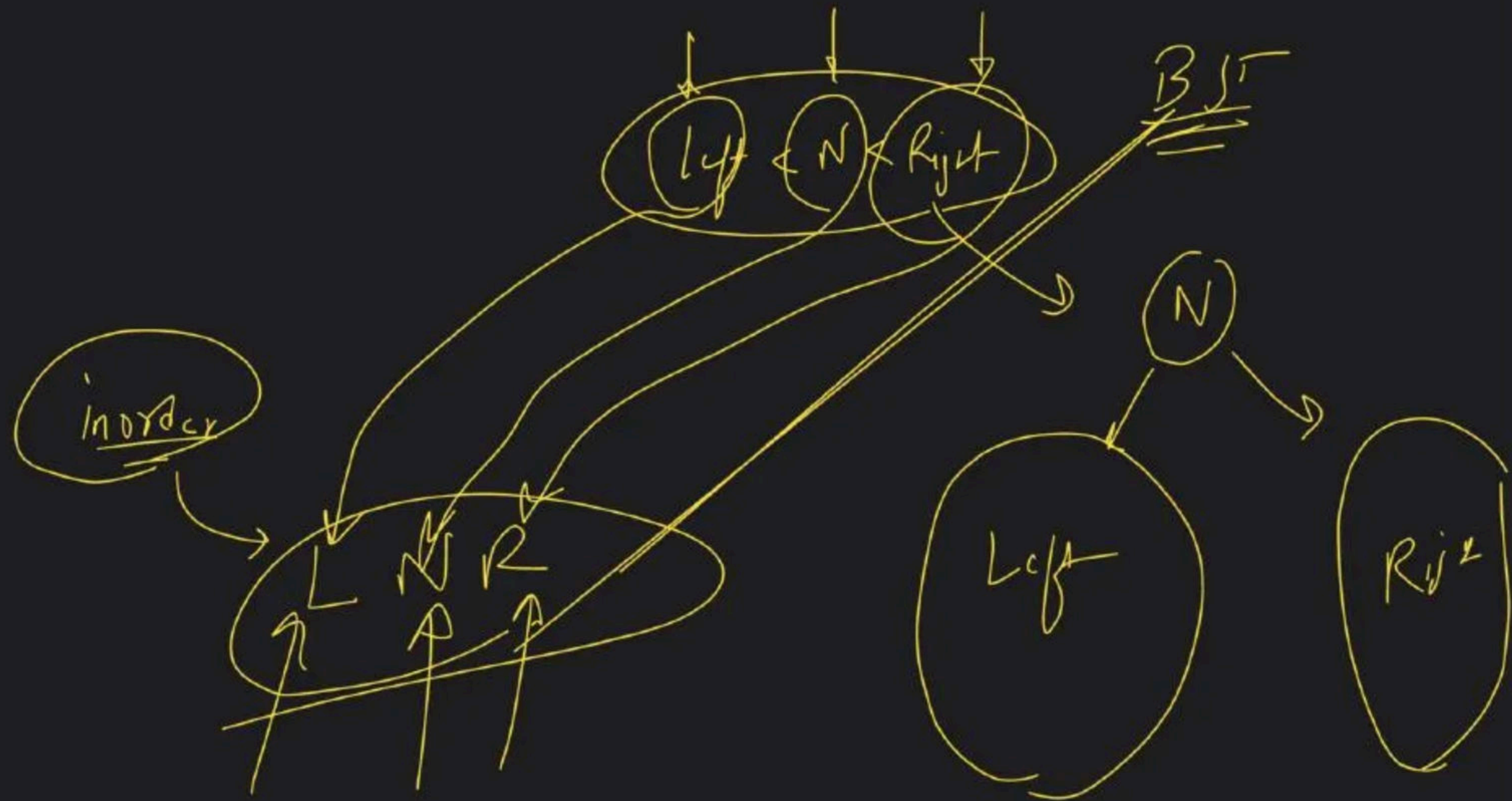
root $\rightarrow$ NULL



1 $\rightarrow$ Rukun jao

100, 50, 200, 20, 70, 250, 150, -1
↑ ↑ ↑ ↑ ↑ ↑ ↑
Rank





Max / Min of BST

root

Minimum

extreme
left

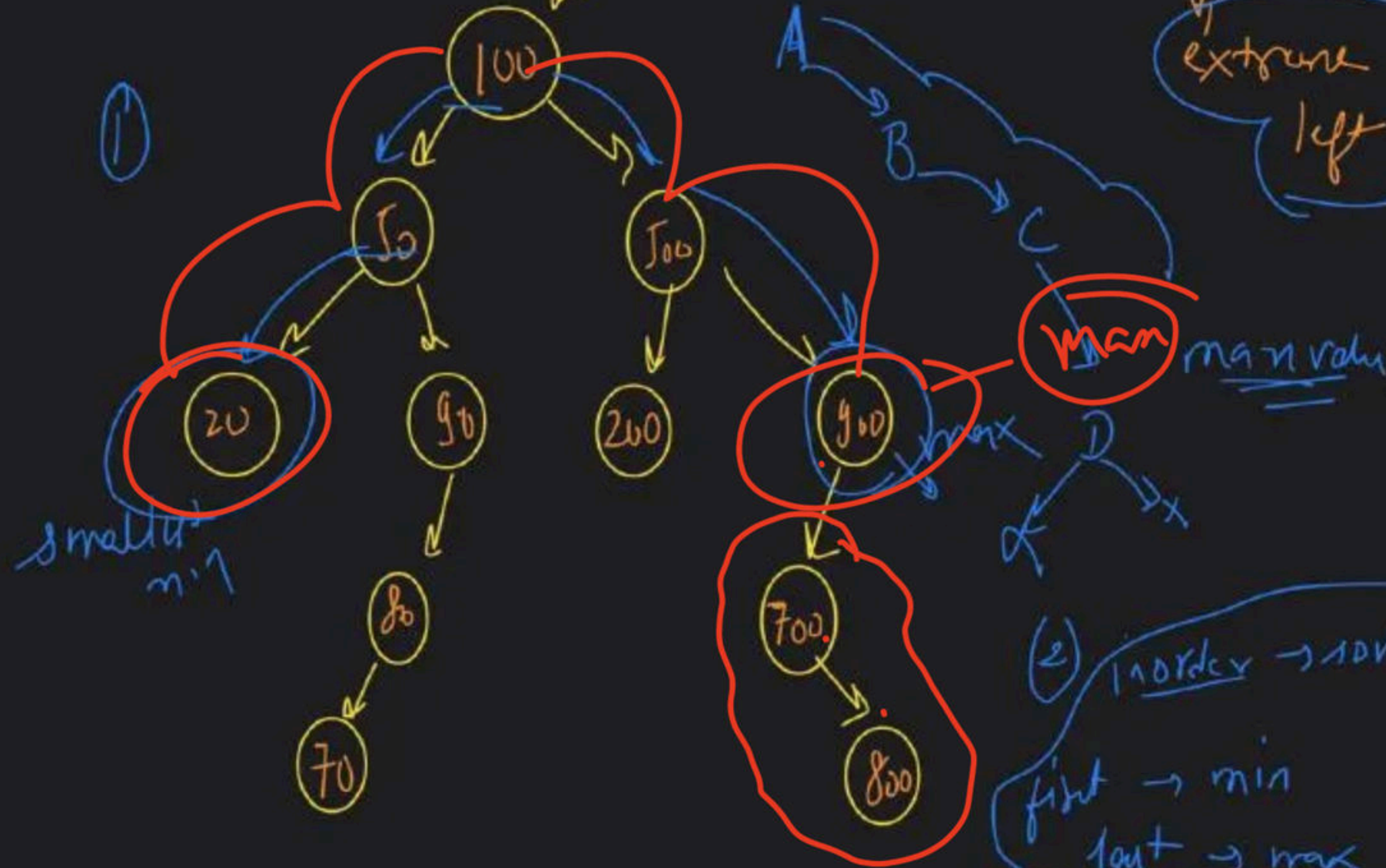
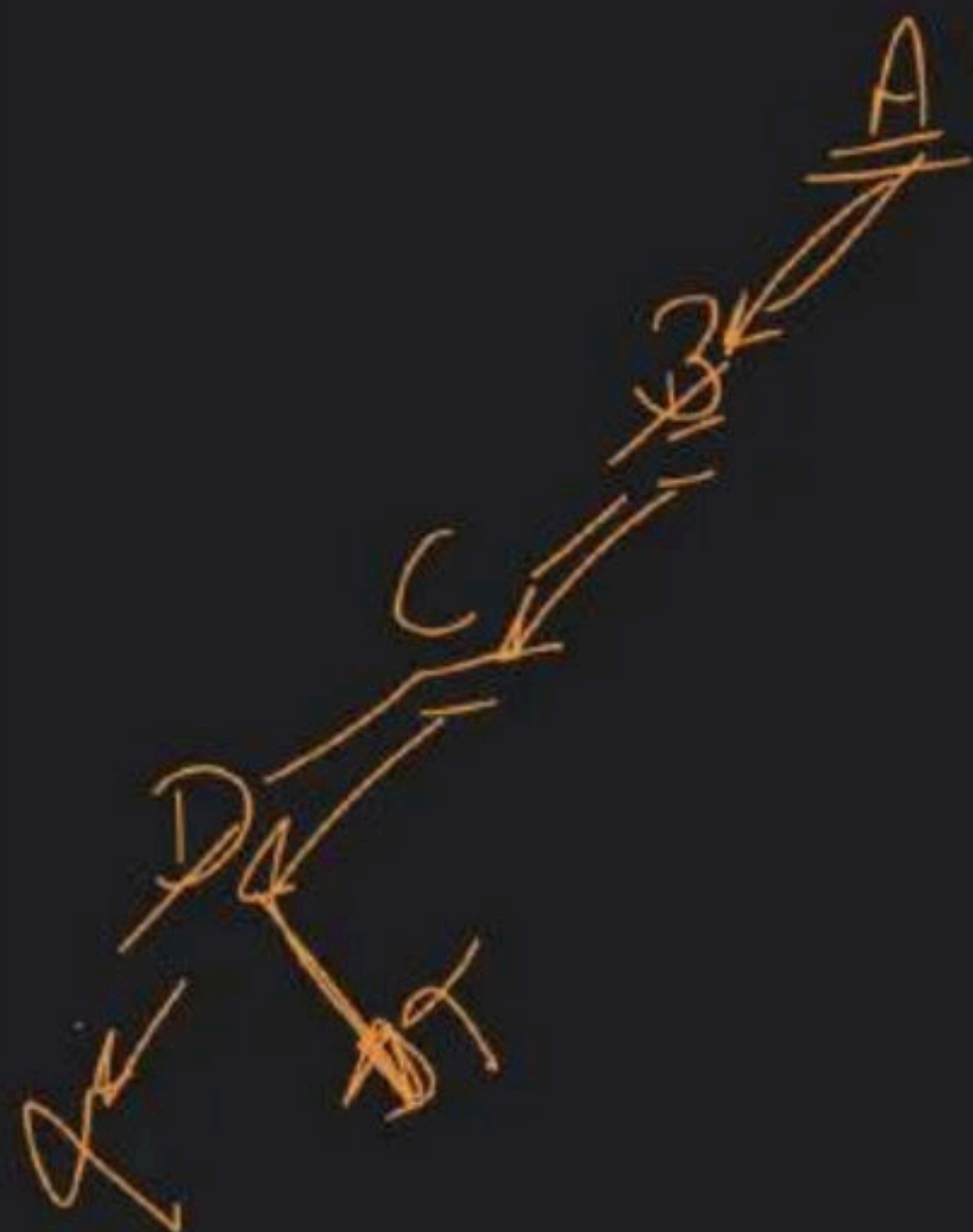
man

man vohu

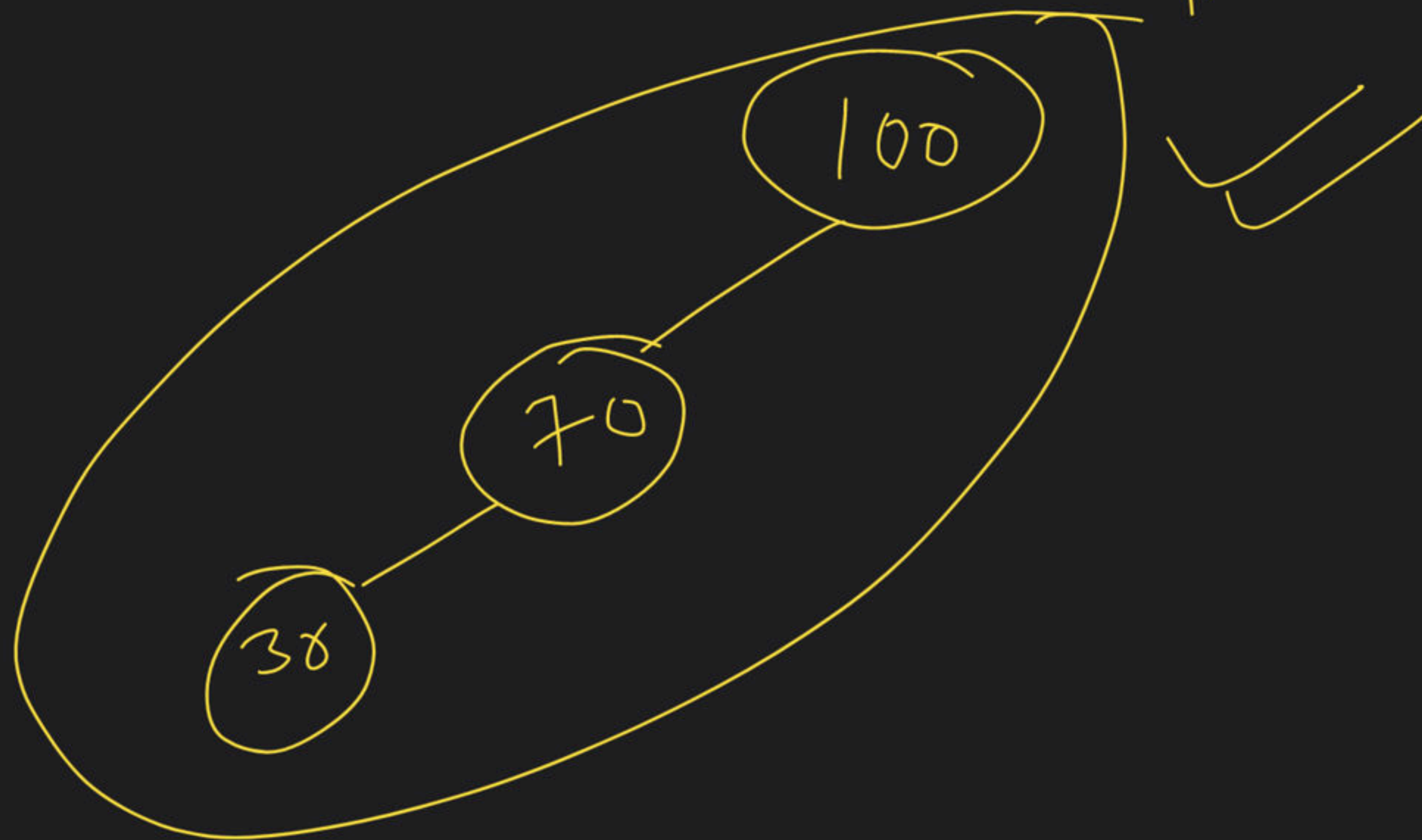
(2) Inorder $\rightarrow$ 1011234

Zeit $\rightarrow$ min

low $\rightarrow$ max

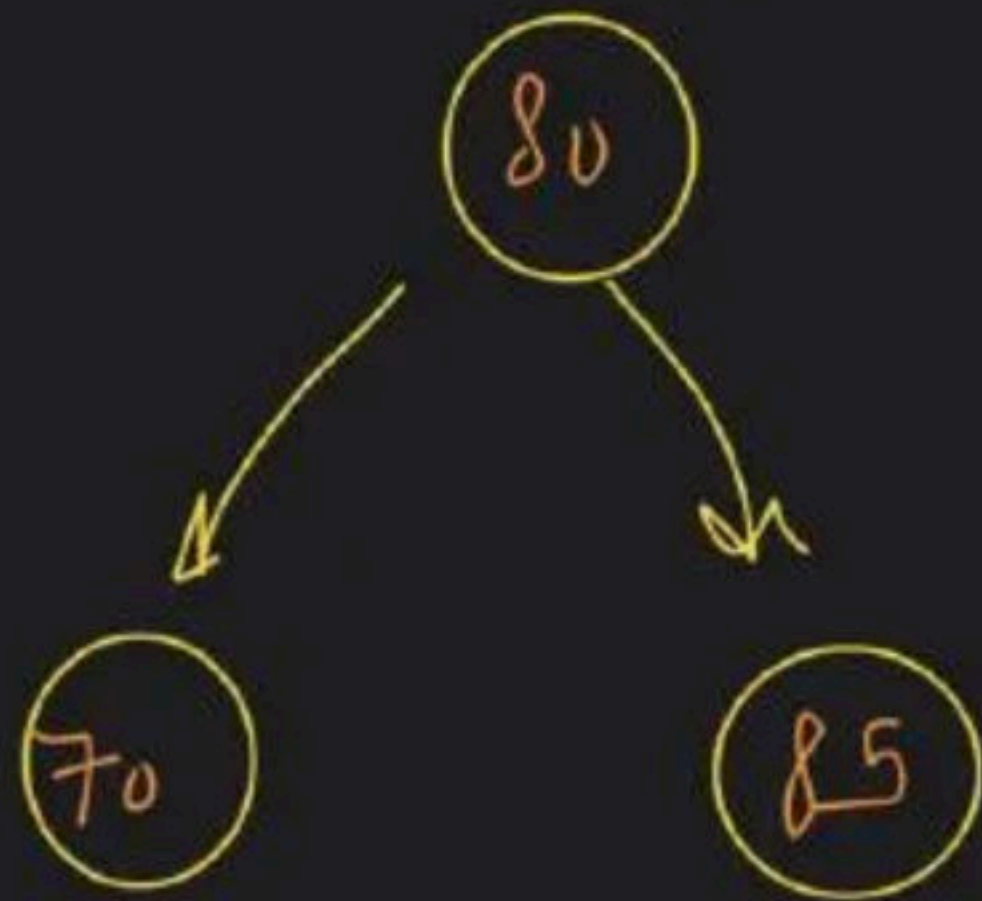


100, 70, 30
↑ ↑

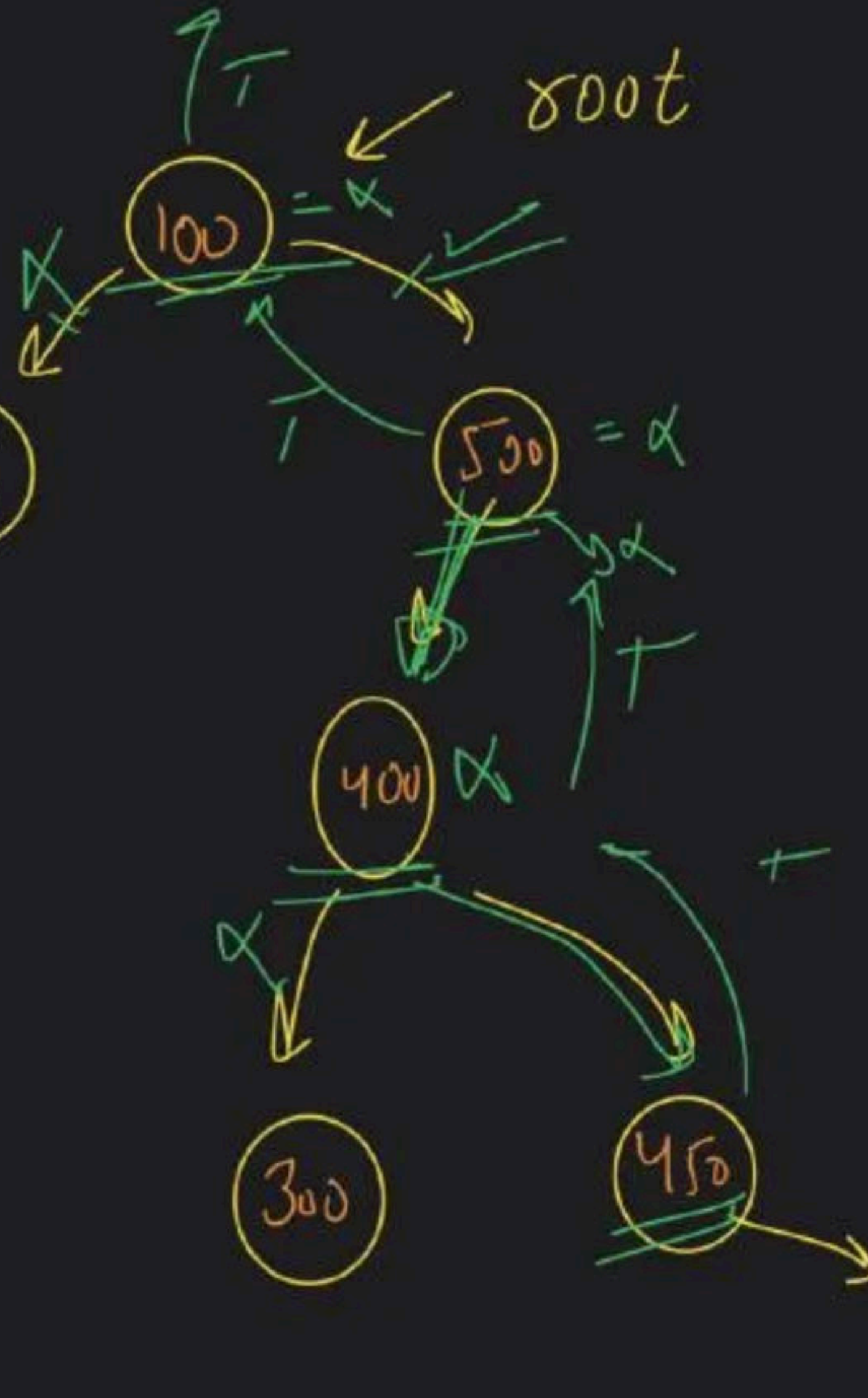


→ Search in BST

$O(n)$ → left right
3.5
↳ $O(\log n)$



90



target = 450

bool → T/F

→ Delete from BST

(1) deletion

(2) BST rakhna
hai

(200) find



100+

100

50

200

300

30

80

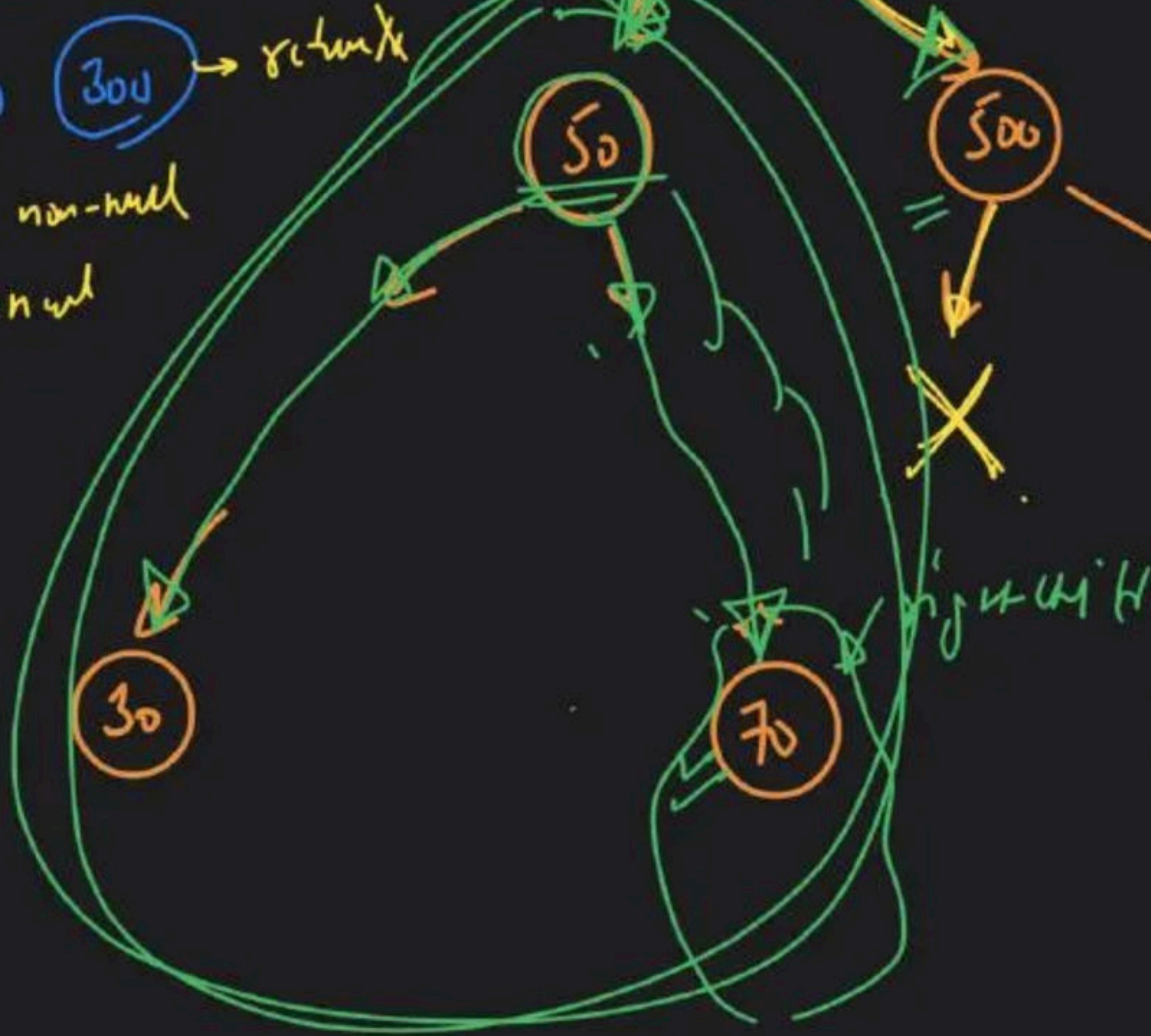
(A)
(B)
(C)
(D)

target = 110

(A) leaf $\rightarrow$ 300 $\rightarrow$ return

(B) left $\rightarrow$ non-null
right $\rightarrow$ null

40



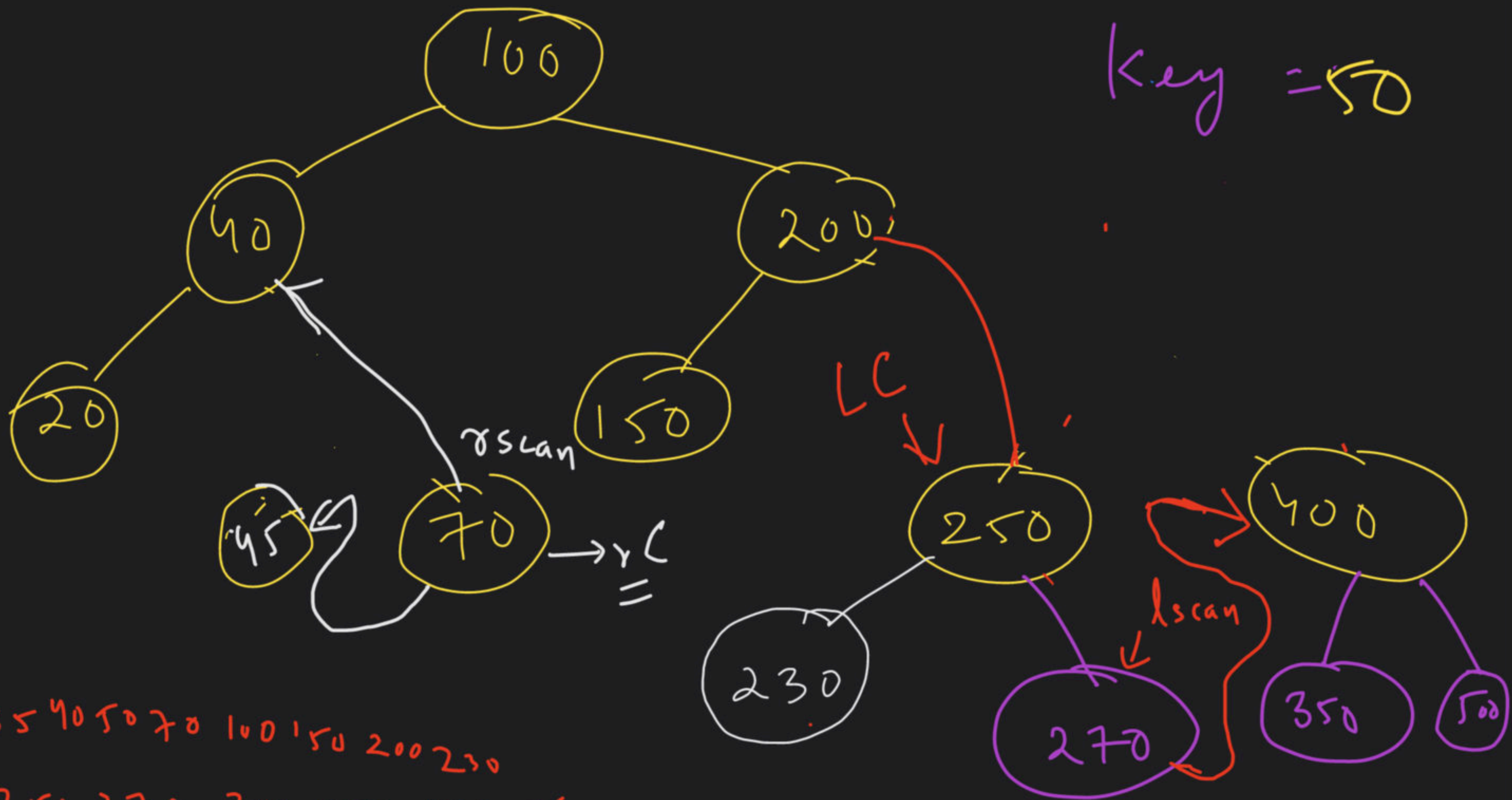
(C) left $\rightarrow$ null
right $\rightarrow$ non null

60

(d) left $\rightarrow$ non null
right $\rightarrow$ non null

100

100 $\rightarrow$ just
check
or
just
bad



20 35 40 50 70 100 150 200 230
 250 270 300 350 400 50

auto rscan = root → right; ①

while (rscan → left)

rscan = rscan → left; ②

step 3: rscan → left = root → left;

rightchild = root → right;

delete root;

return RC;

① Find the node

② Case Basis deletion

↳

①

Leaf node

②

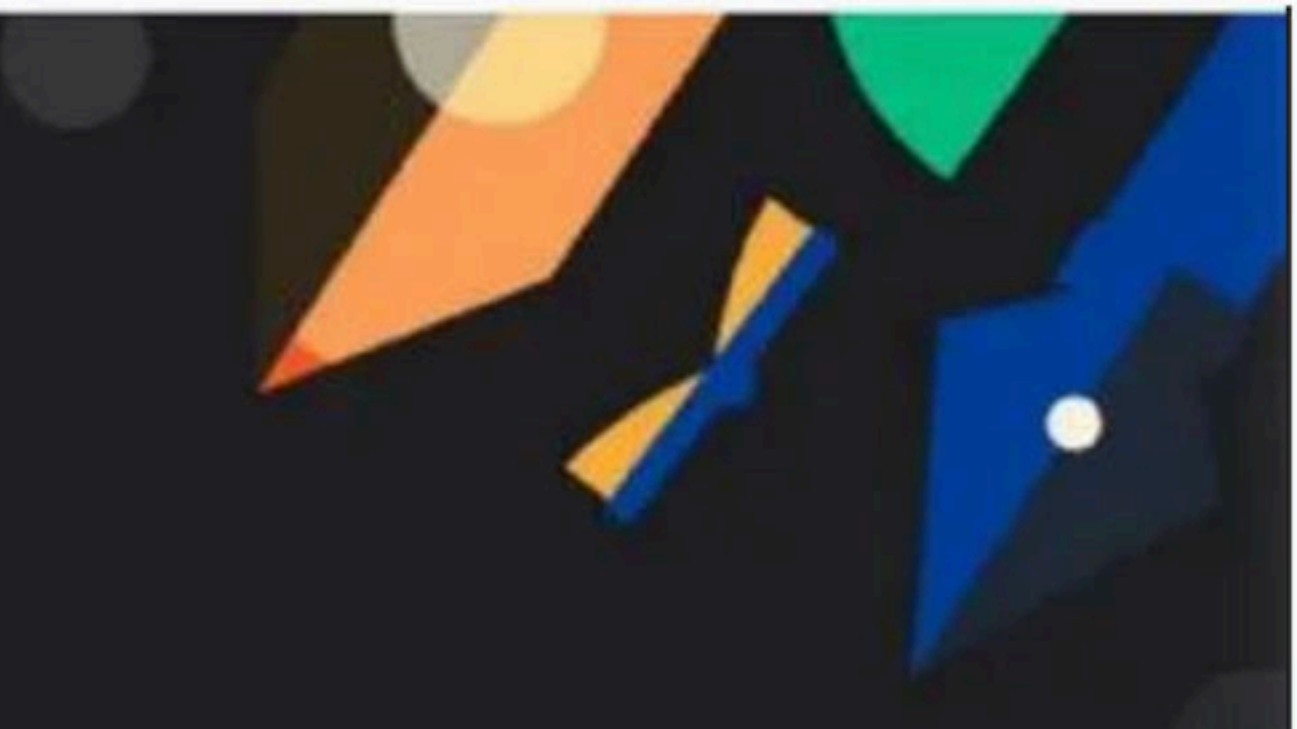
left child only

③

Right " "

④

Both " exist

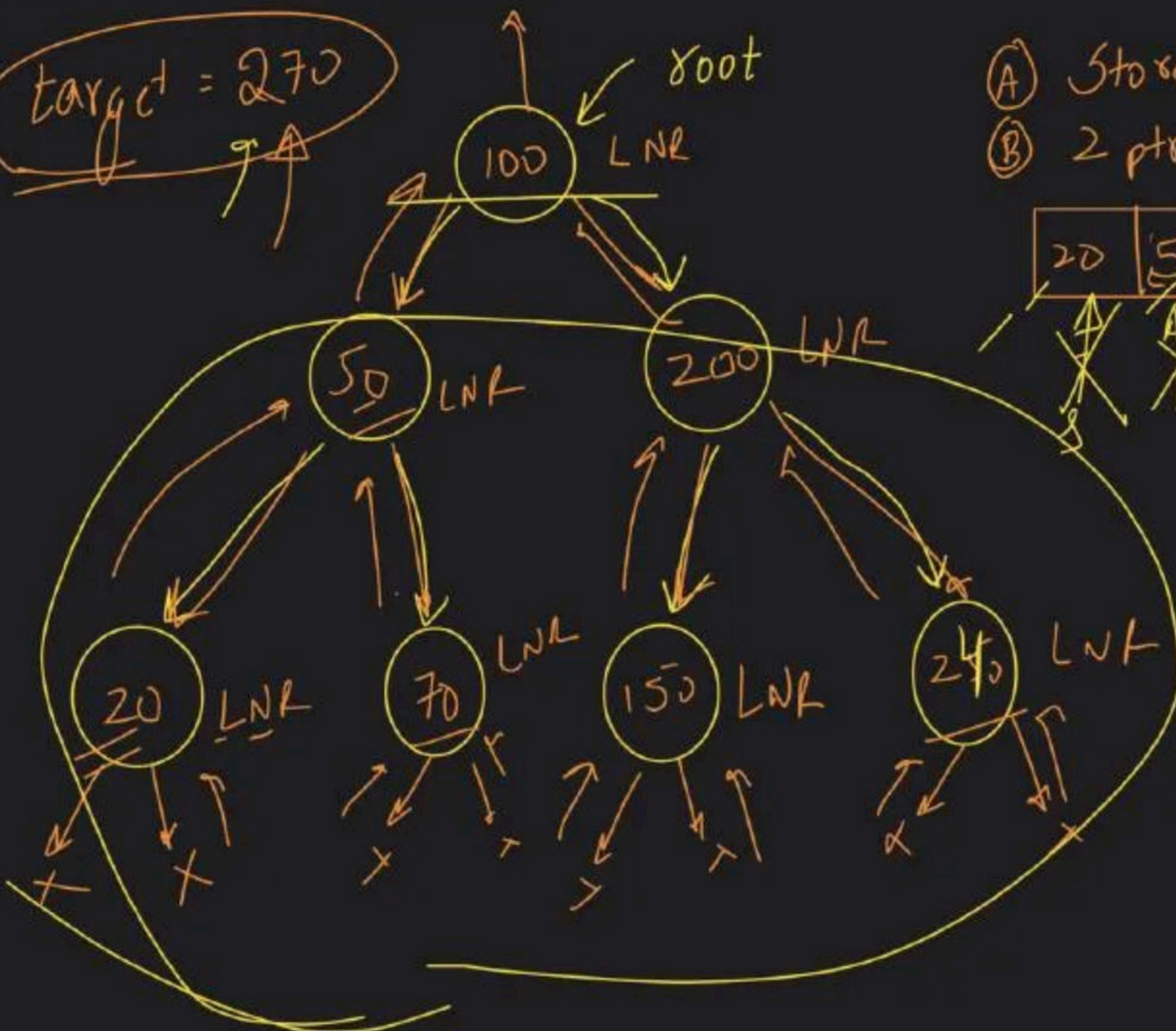


Binary Search Trees Class - 2

Special class

target = 270

- (A) Store Inorder
- (B) 2 ptr approach



20	50	70	100	150	200	240
----	----	----	-----	-----	-----	-----

$250 < 270$

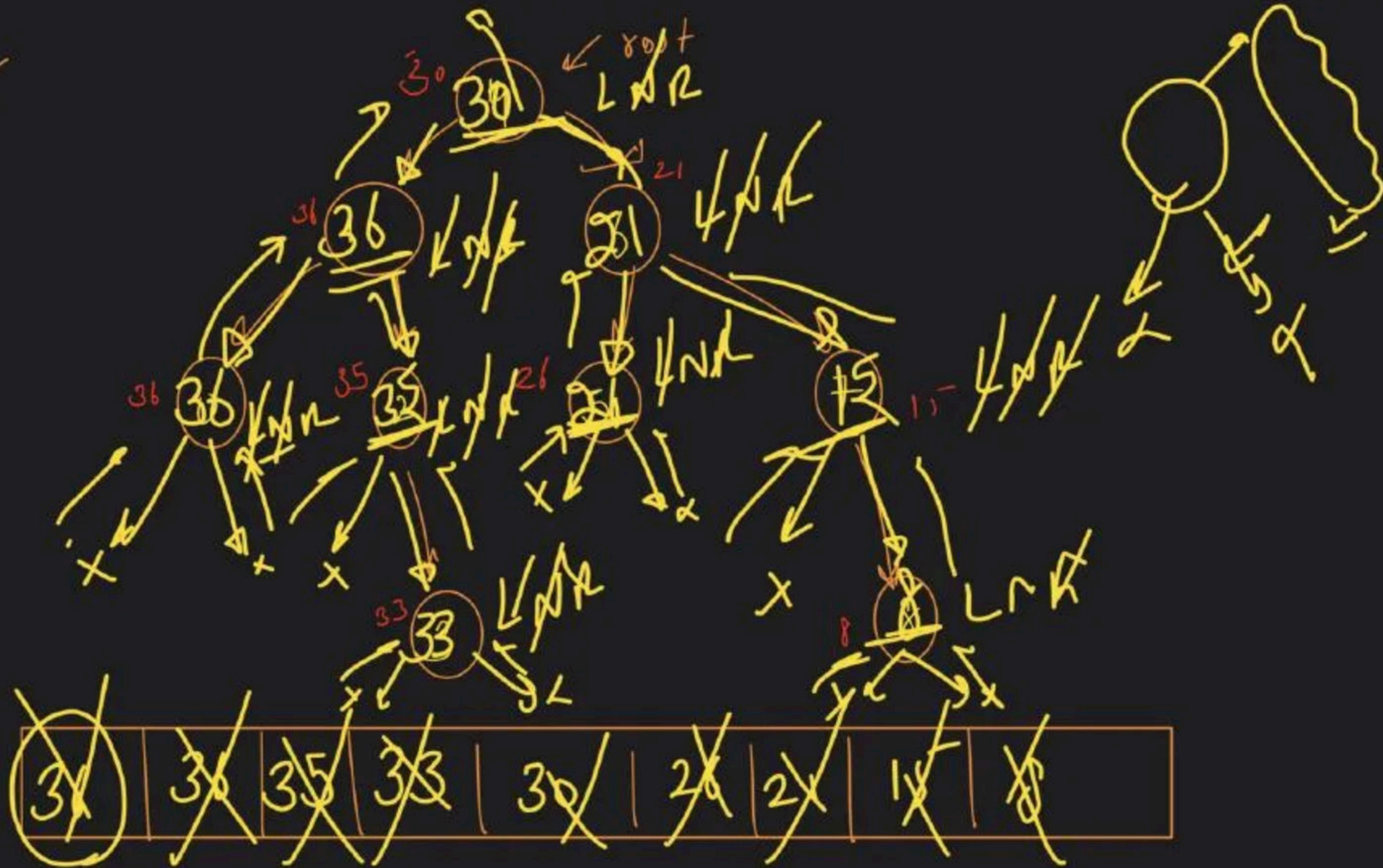
$290 < 270$

$sum < target \rightarrow ++$

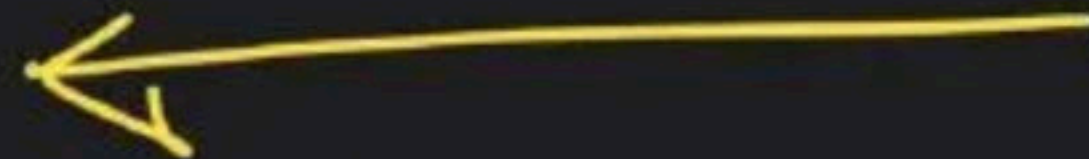
$290 > 270$

$sum > target \rightarrow --$

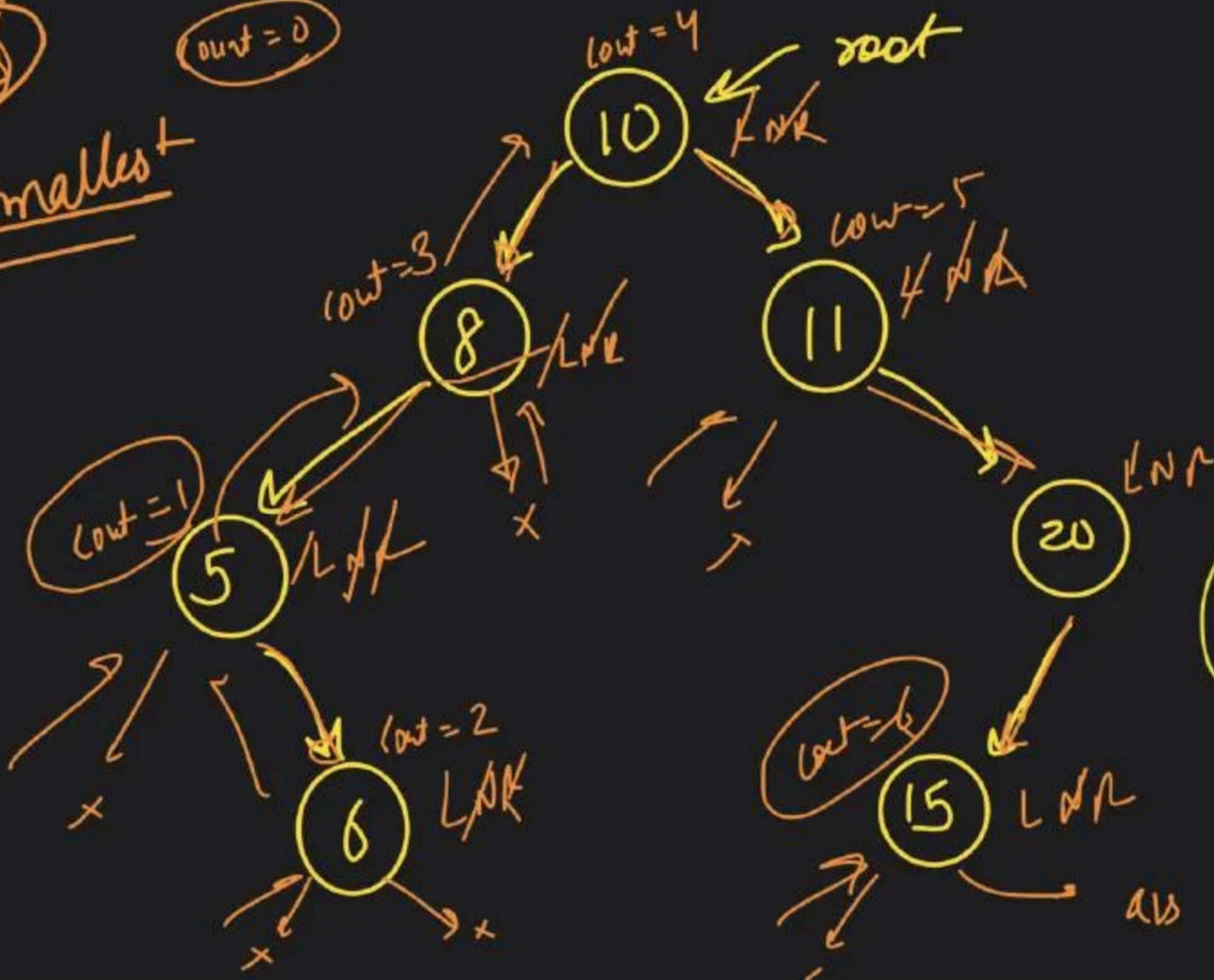
(A) store
in order



30	26	35	32	30	²⁶ 5	²¹ 1	¹⁵ 5	<u>8</u>
----	----	----	----	---------------	----------------------------	----------------------------	----------------------------	----------



$K = 4$
 6^{th} smallest
 $count = 0$



inorder
LNR

NLR
 preorder

→ LCA

①

BST

$O(n)$

$O(n)$

$O(n)$ → L
→ R

BST →

$O(n)$

$O(\log n)$

$O(\log n)$ → L
or
R

Cases

(A)



Left

P19

sub-tree



③

root

right

left

right

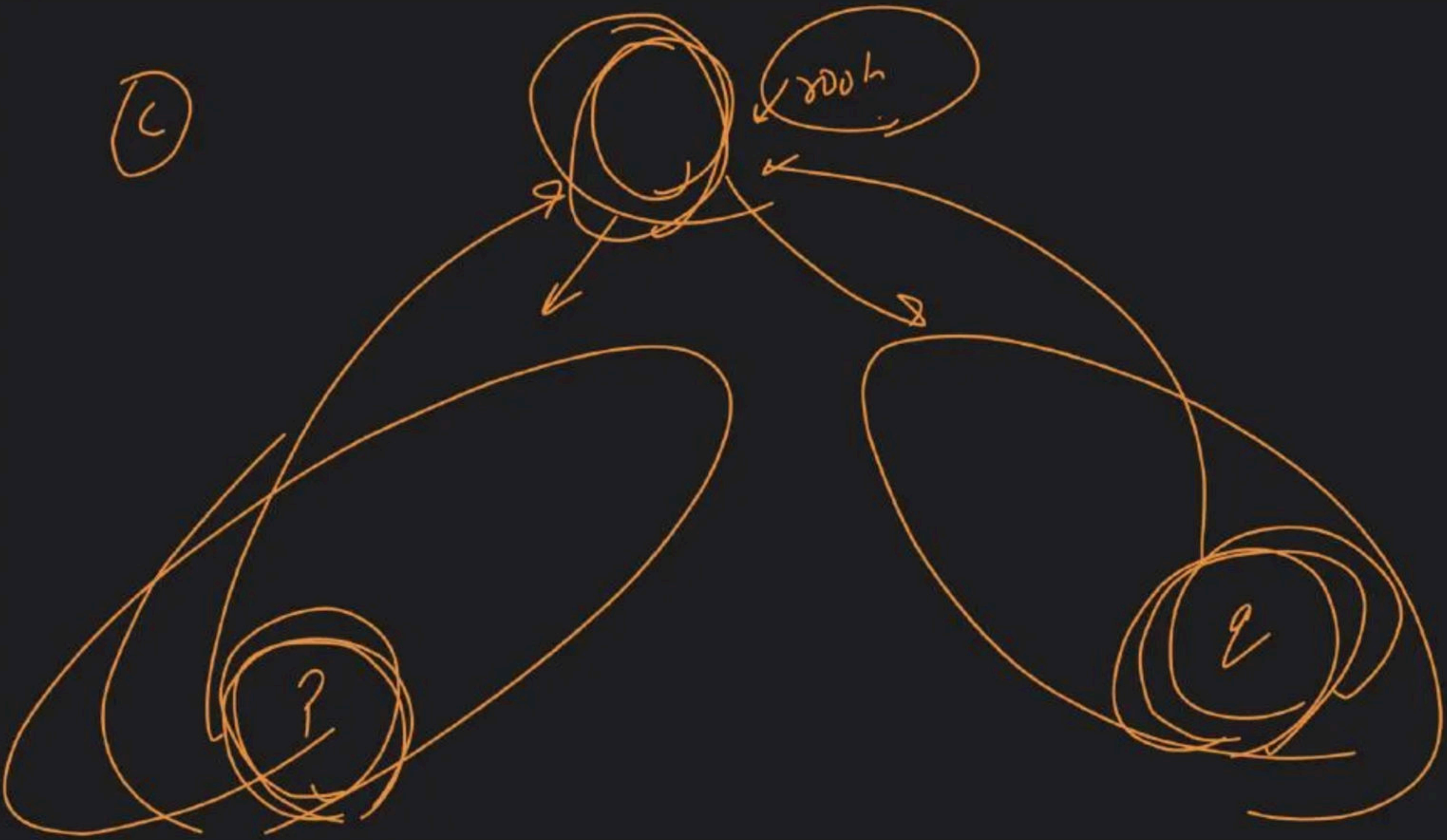
①

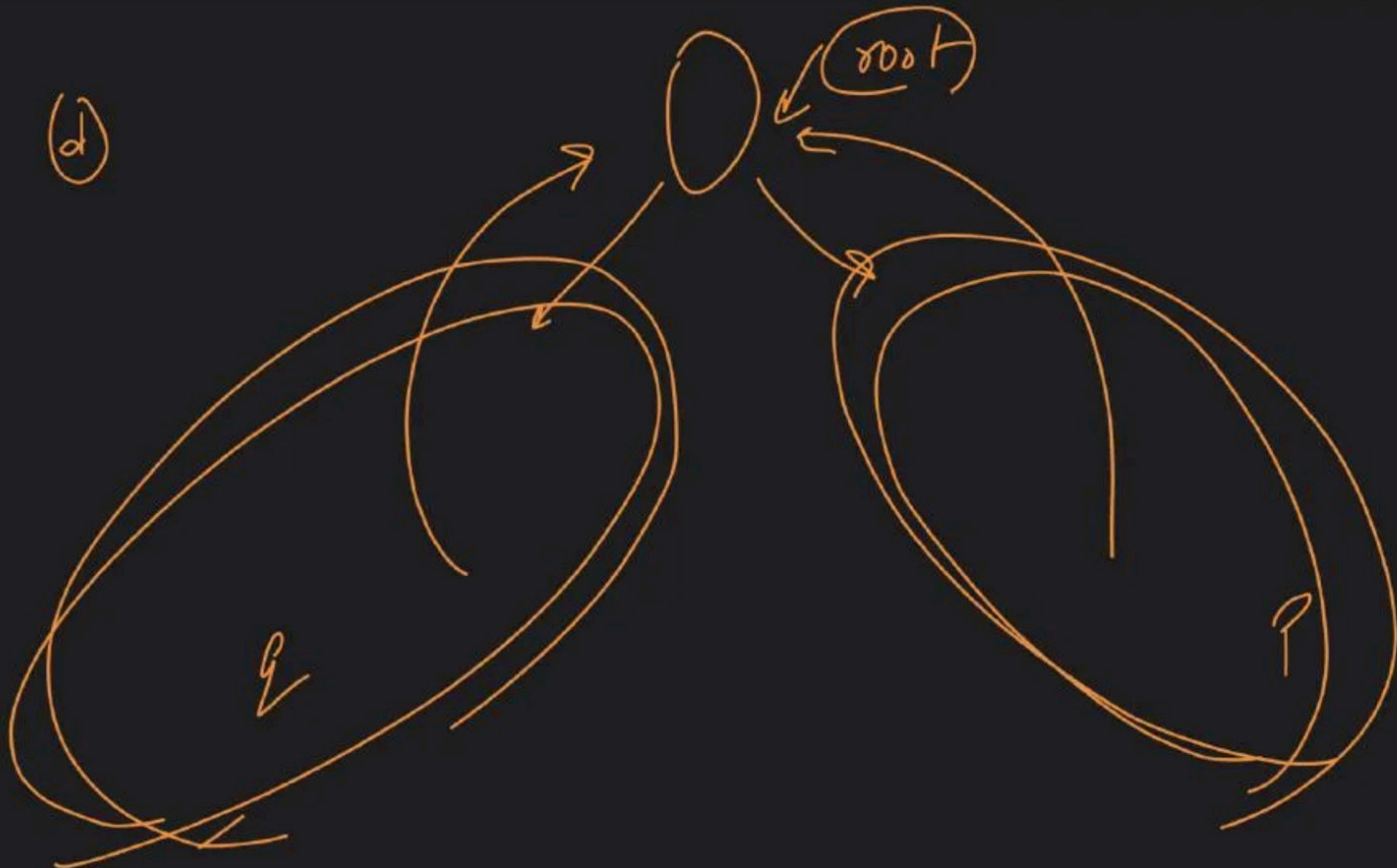
400h

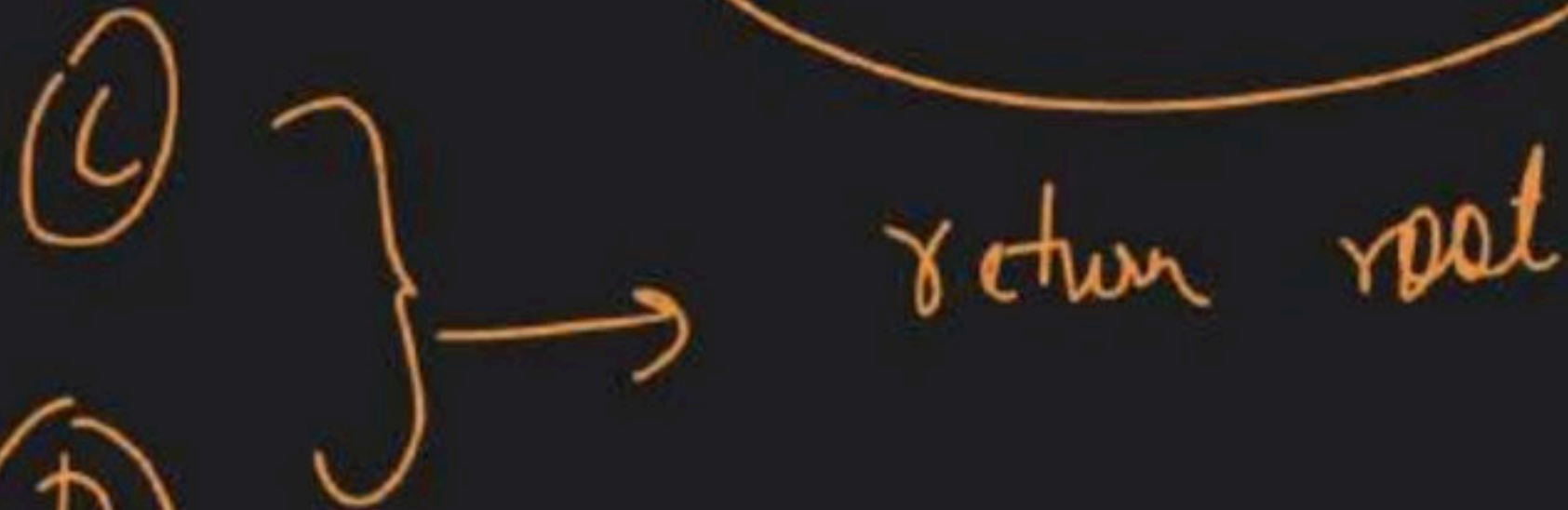
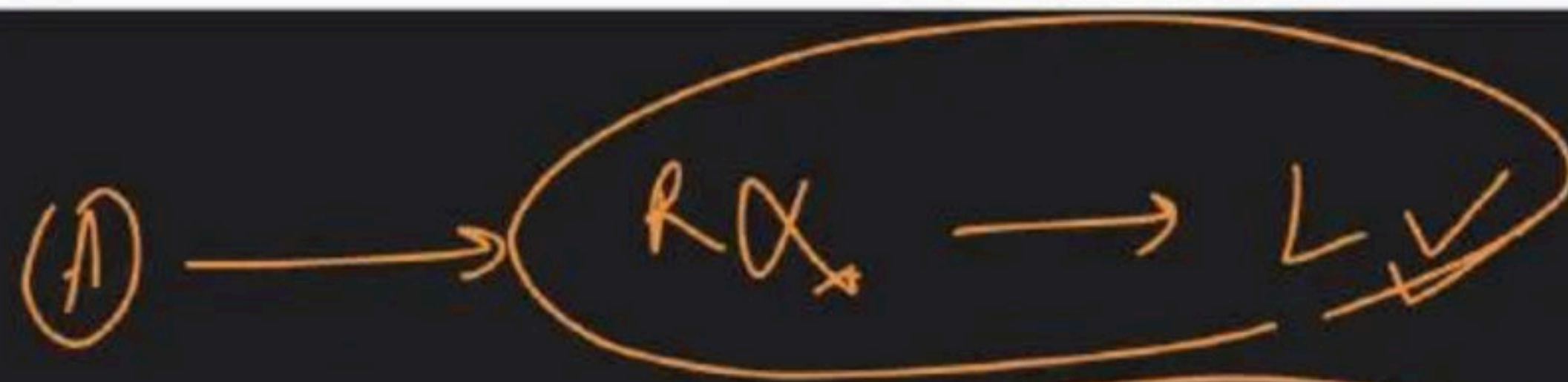
1

2

?







L & R
 α

L or R

LC $\Rightarrow$ 538

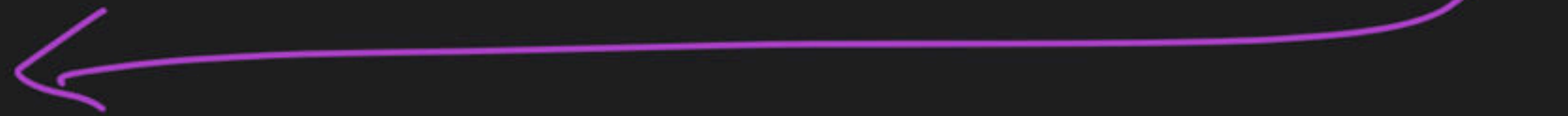
M1

① inorder store

② prefix sum

③ update Tree

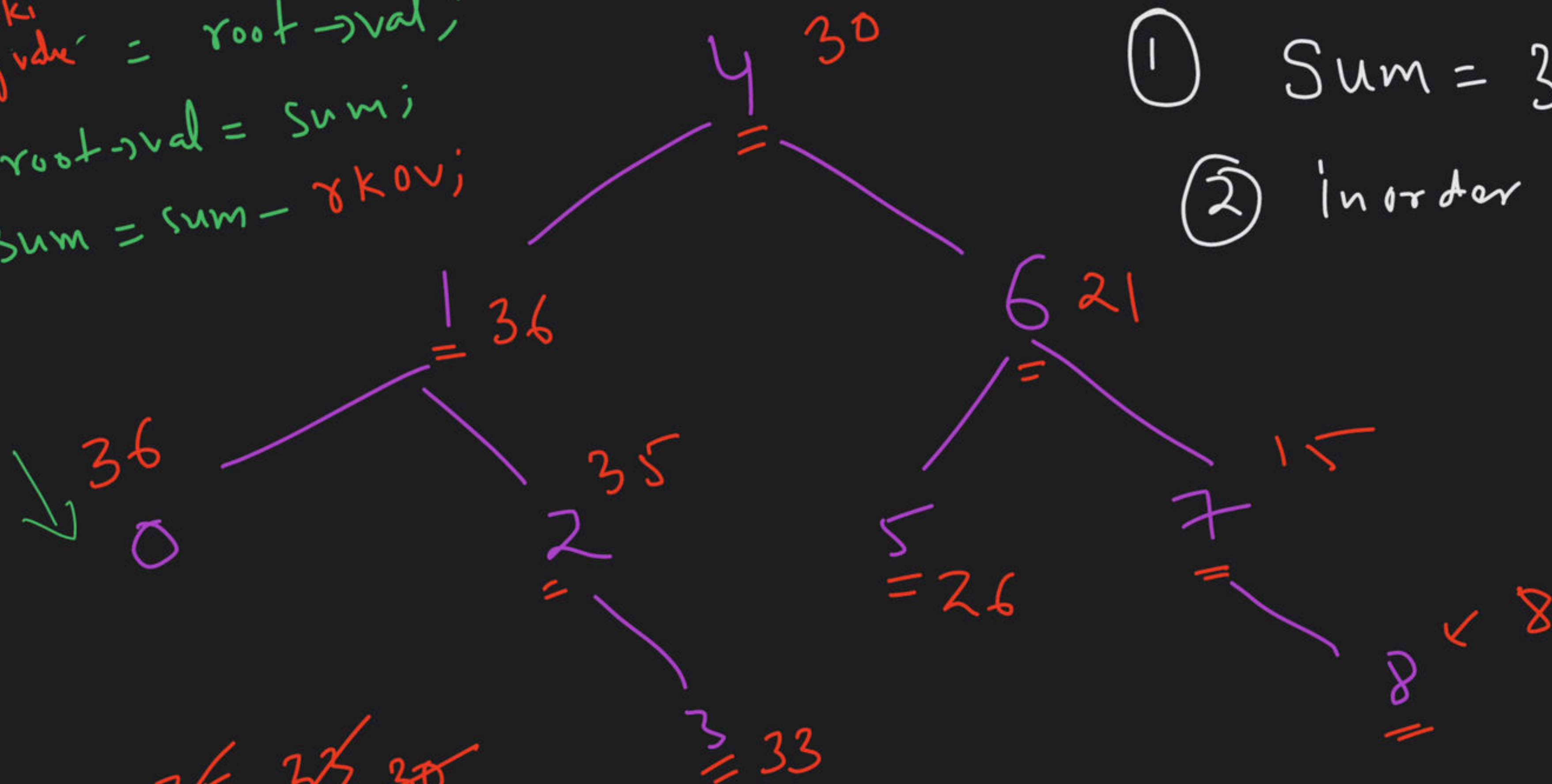
SC: $O(n)$



$\text{rootki} = \text{root} \rightarrow \text{val};$
 $\text{root} \rightarrow \text{val} = \text{sum};$
 $\text{sum} = \text{sum} - \text{rootki};$

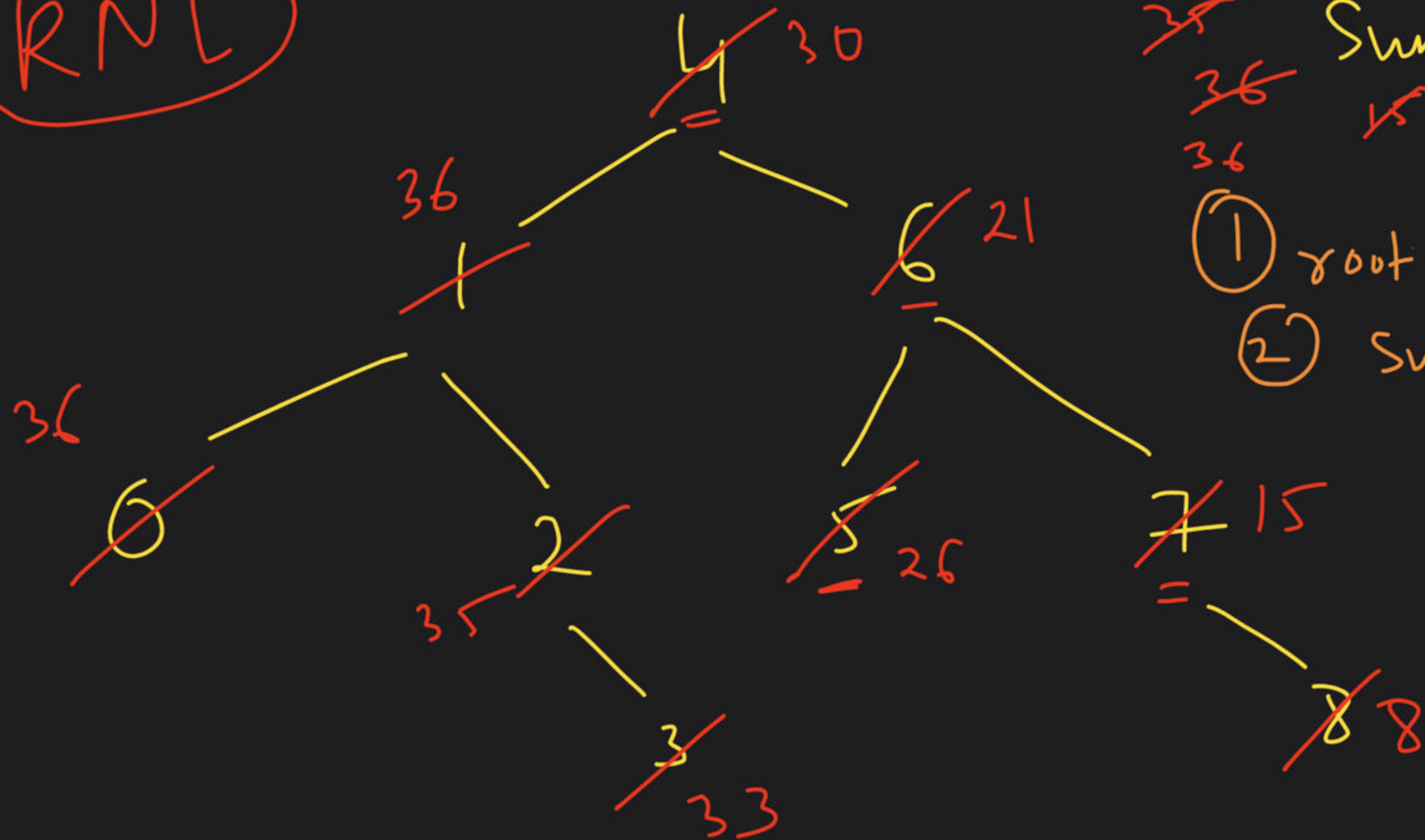
① $\text{Sum} = 36$

② in order traversal



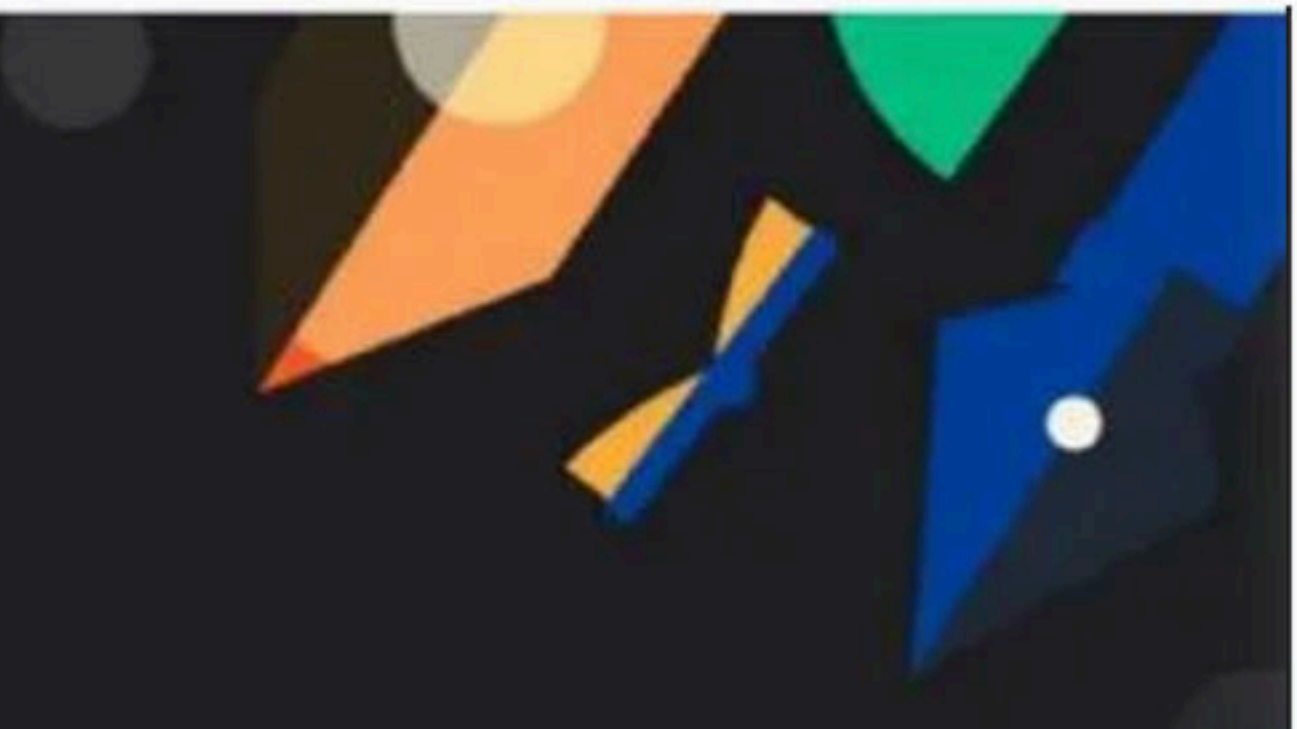
$\text{Sum} = \cancel{36} \ \cancel{35} \ \cancel{30} \ \cancel{26} \ \cancel{21} \ \cancel{15} \ \cancel{8} \ 0$

RNL



~~35~~
~~36~~
~~36~~
Sum = ~~33~~
36
15 21 26

① root $\rightarrow$ val += sum
② Sum = root $\rightarrow$ val



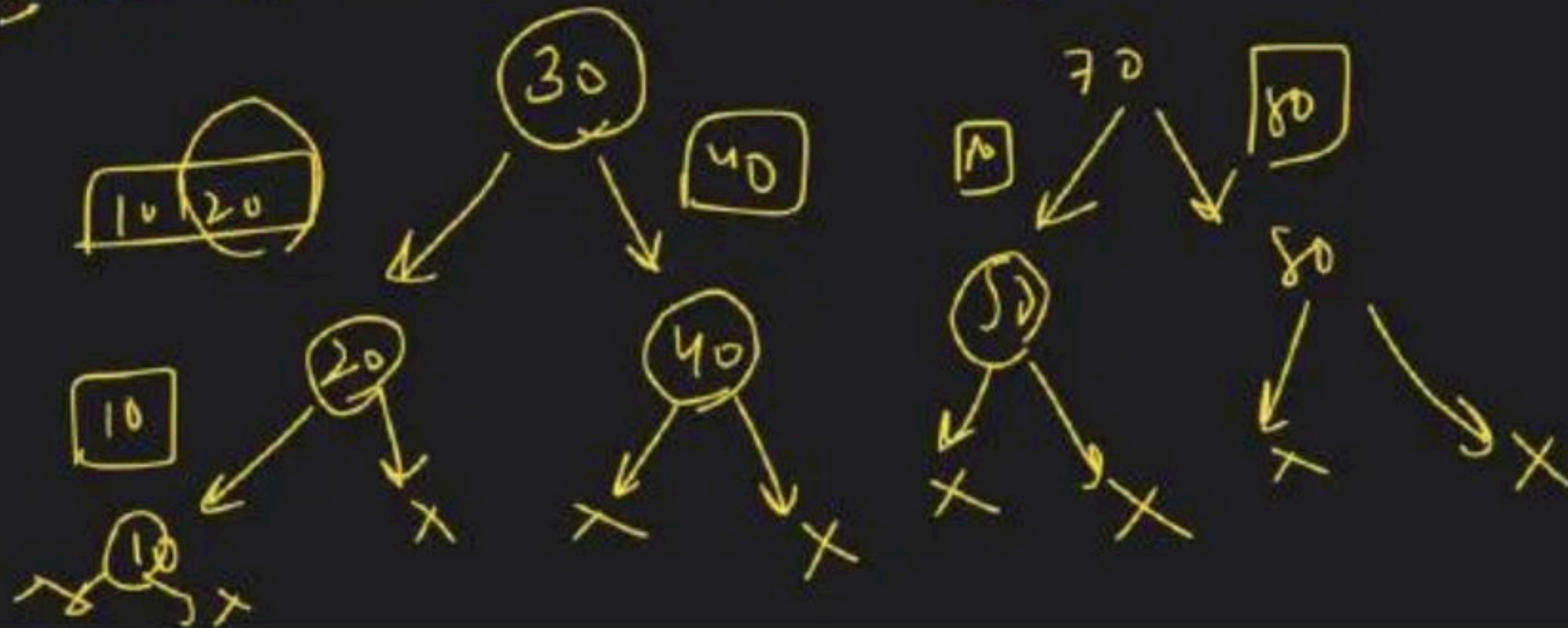
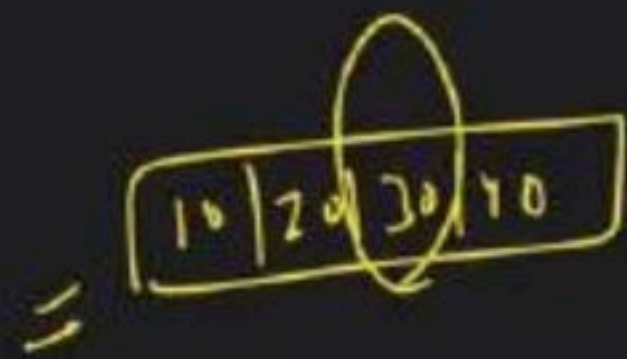
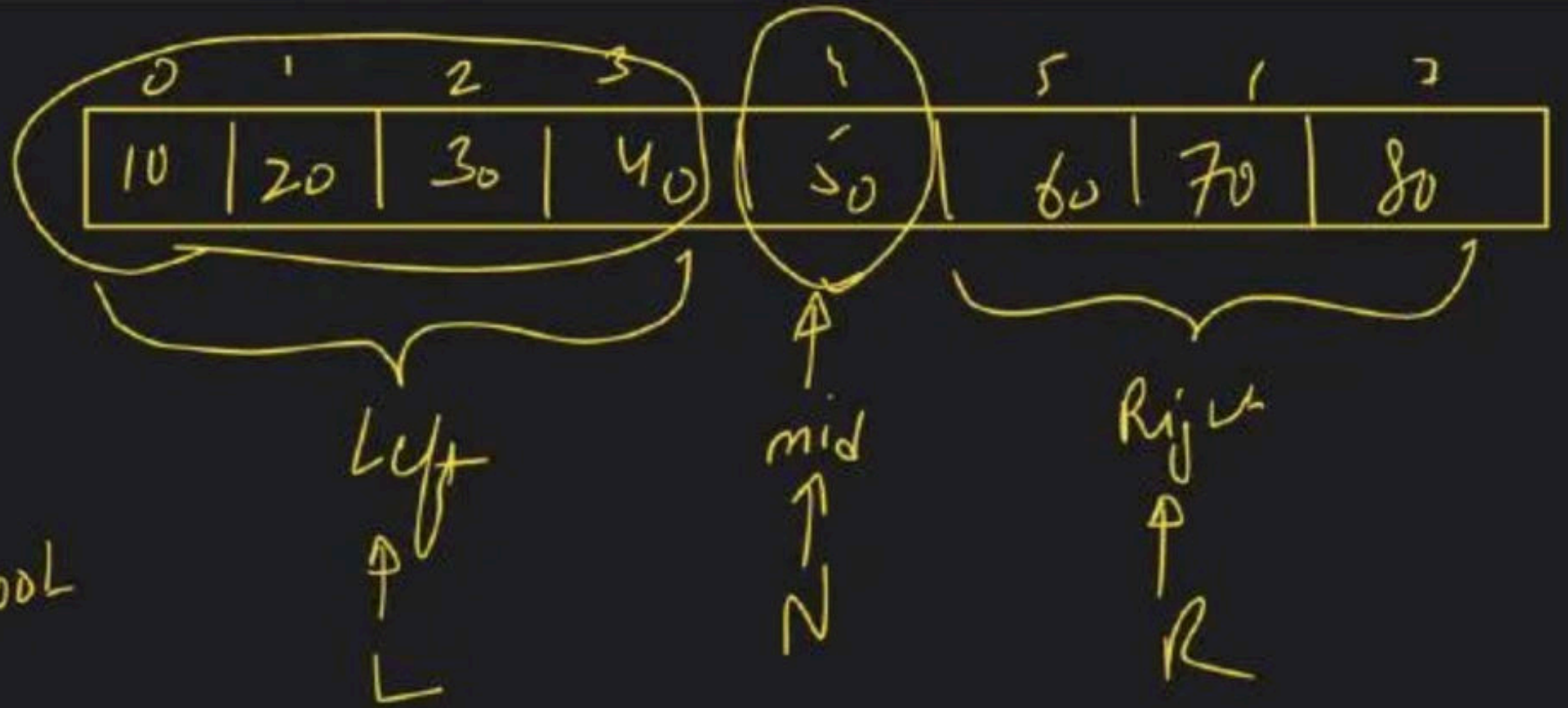
Binary Search Trees Class - 3

Special class

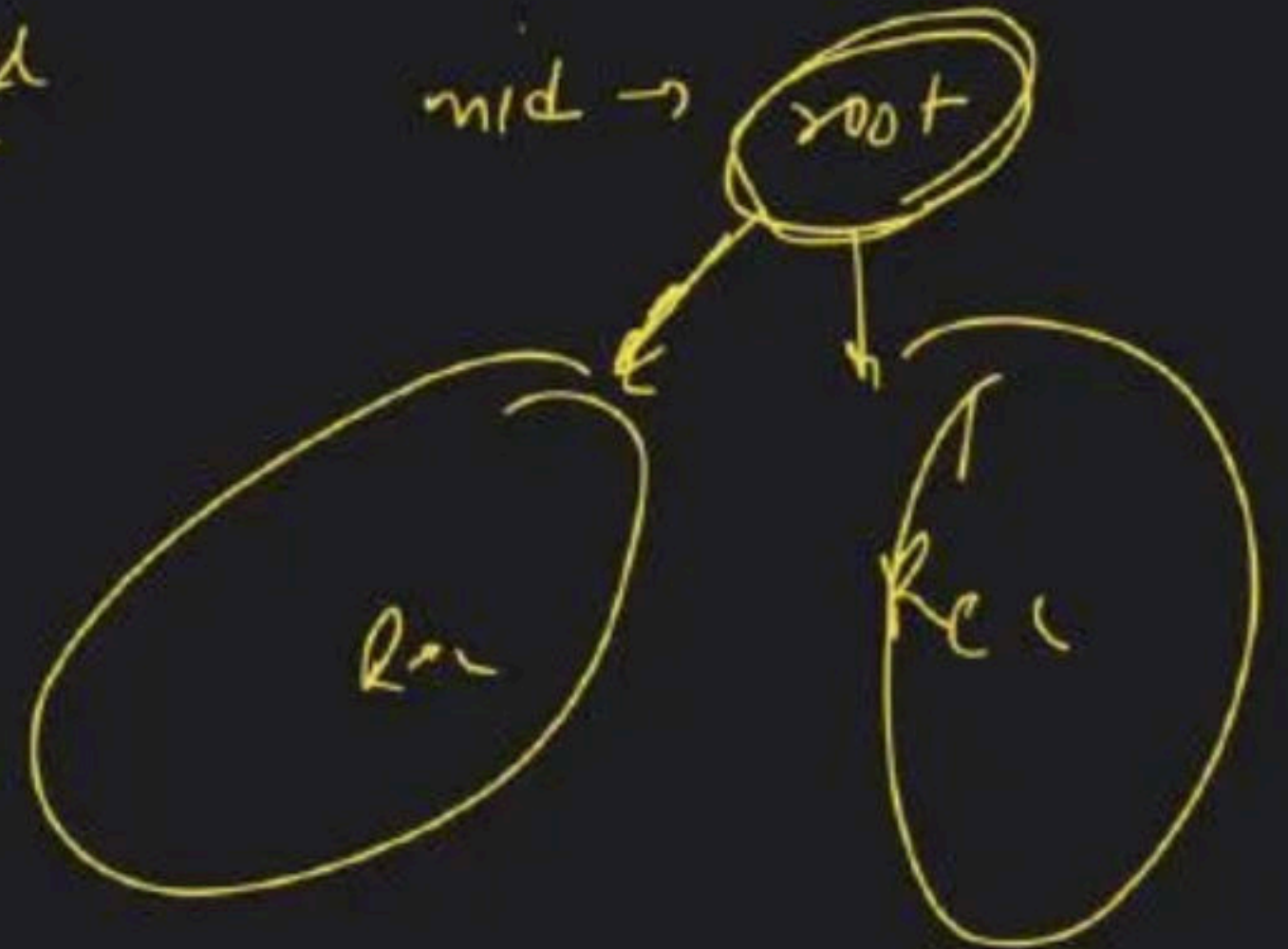
① BT $\xrightarrow{\quad}$ construct BT using In/Pre/Post

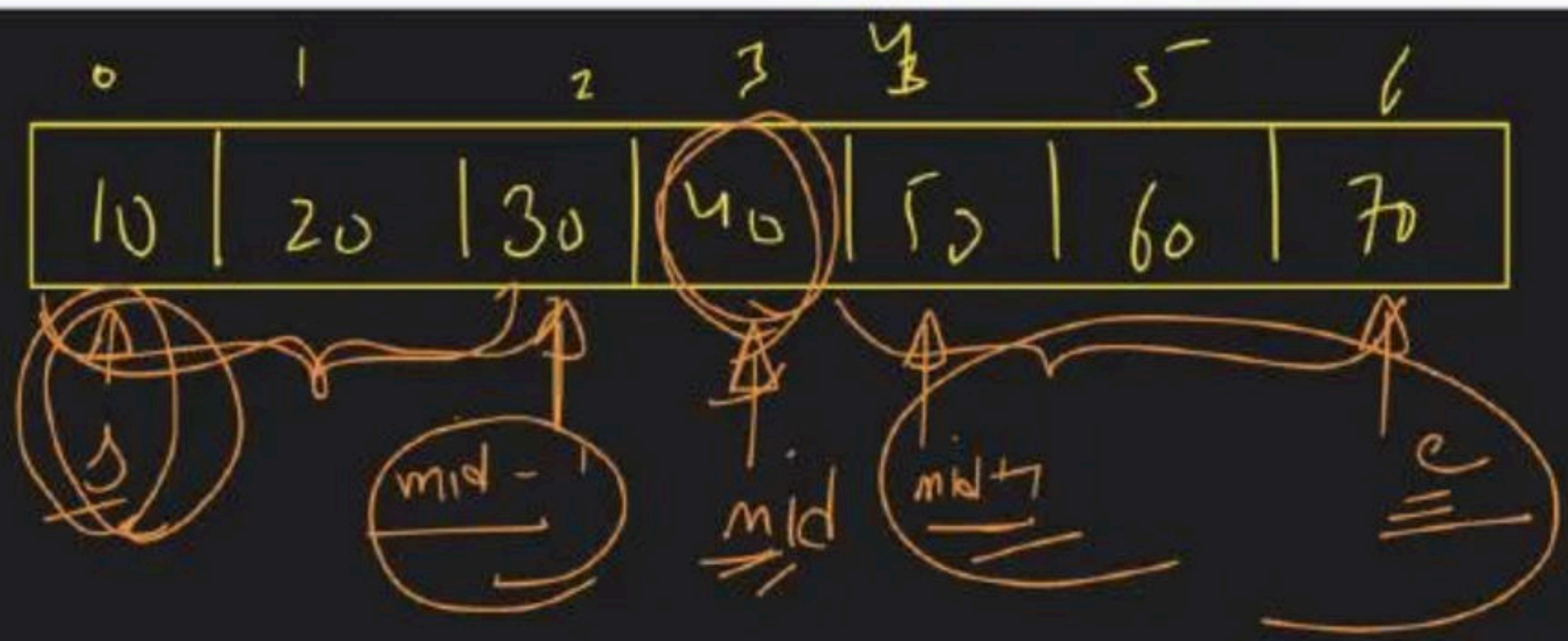
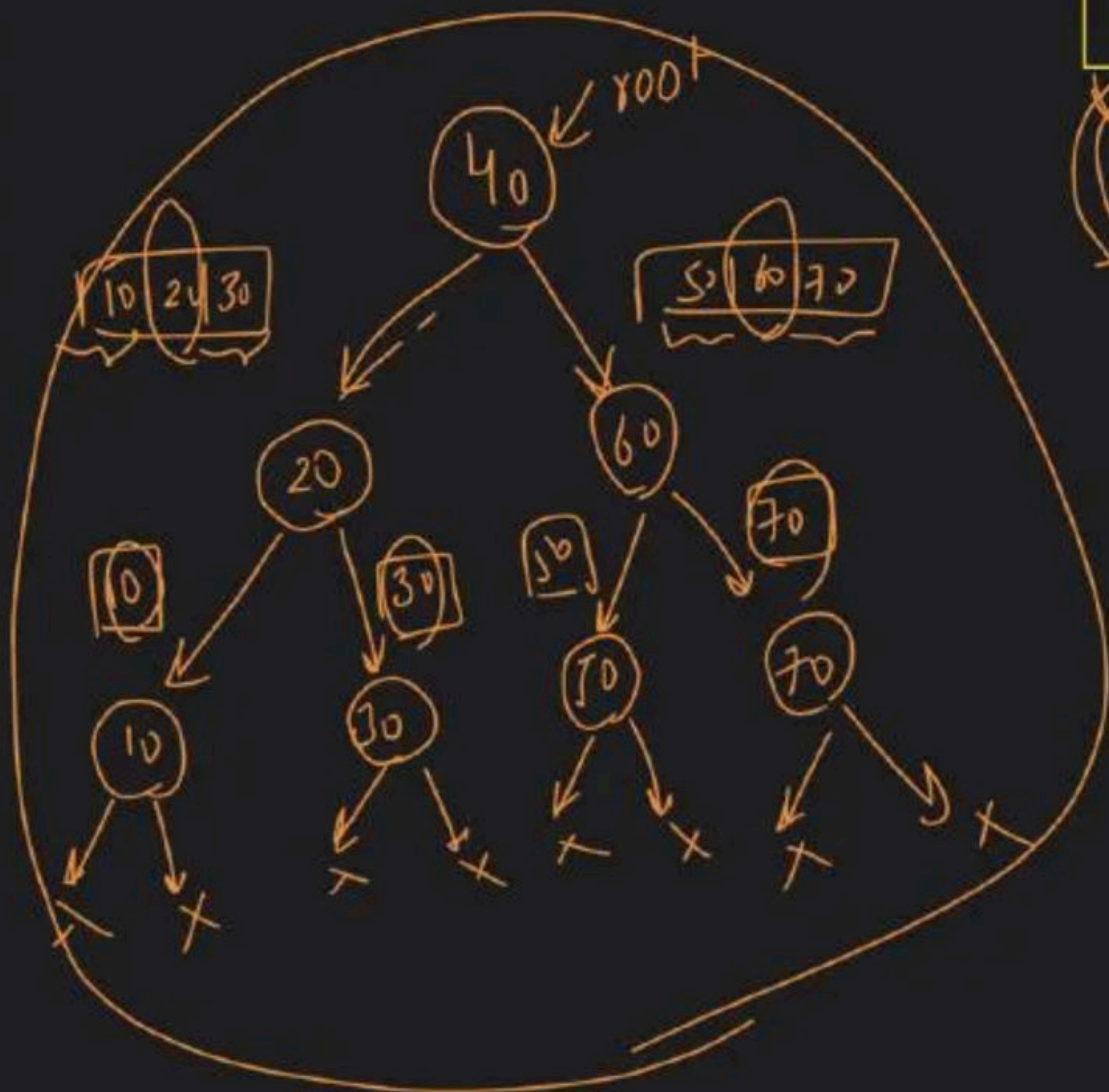
BST $\xrightarrow{\quad}$ create BST using Inorder/
Sorted Array

Inorder
↓
LNR



1 case





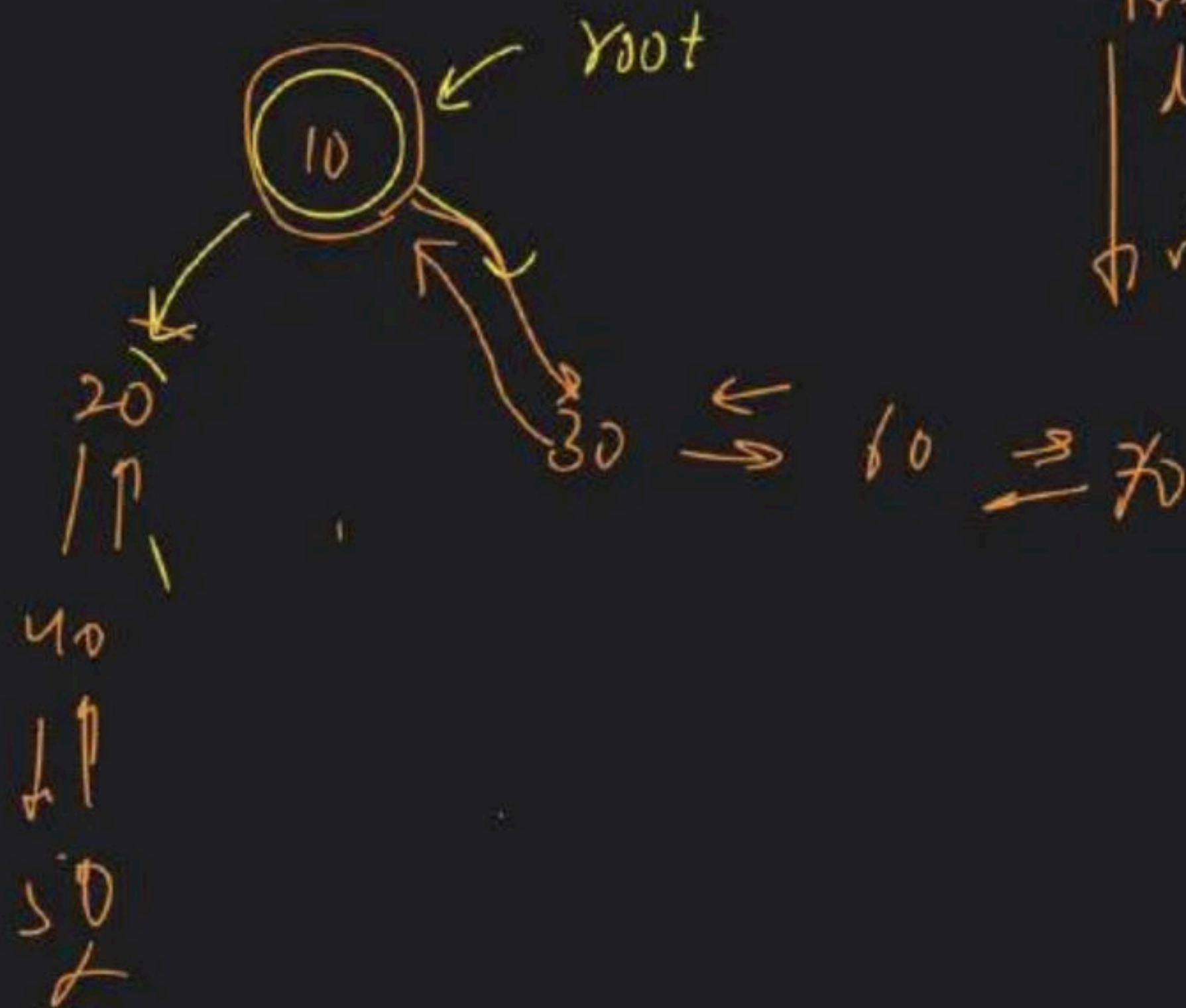
→ BT → DLL → sorted

BST → DLI → $\frac{x}{p}$
↳ sorted

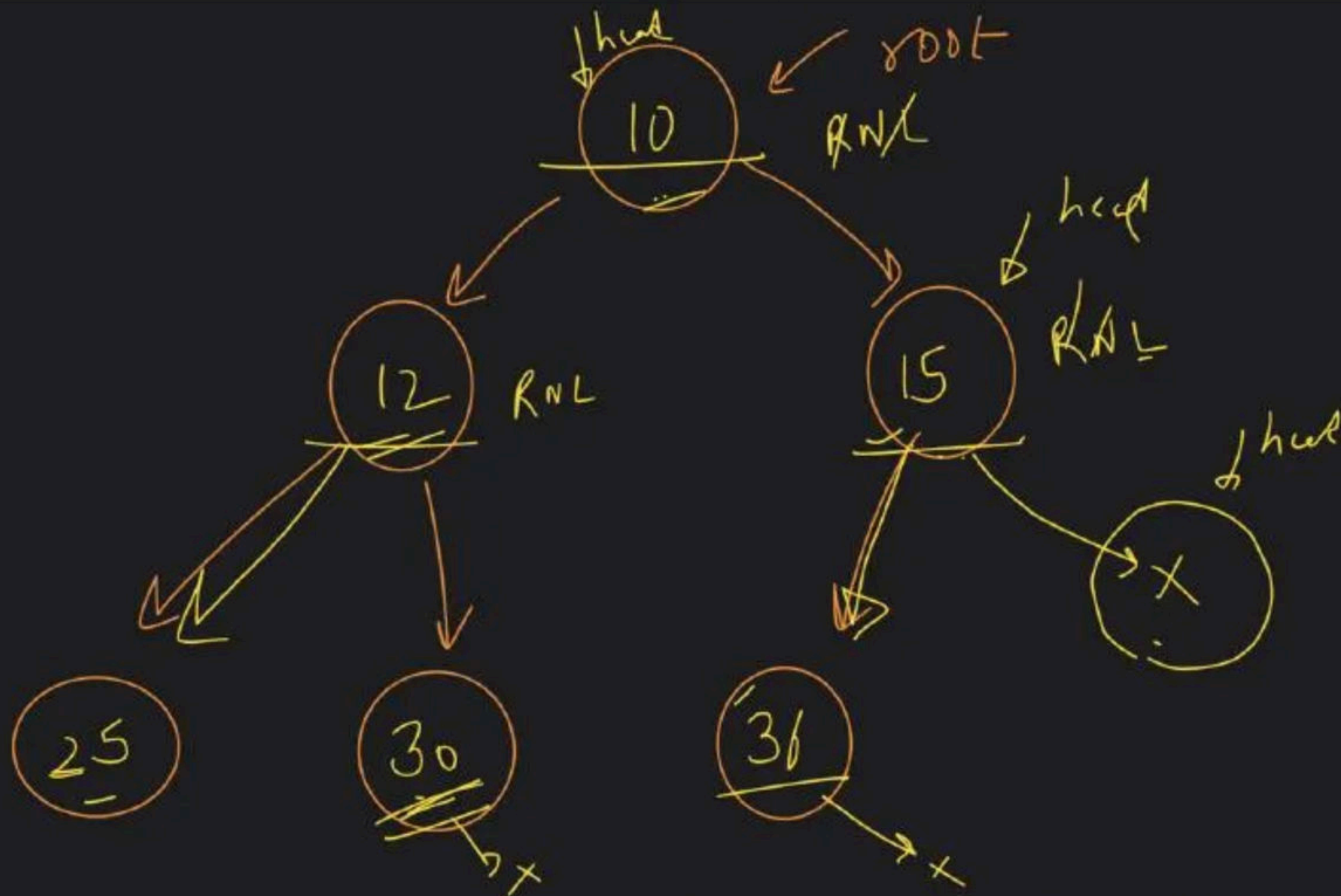
→ BI to DLL



in → LNR
pre → NLR
post → LNR

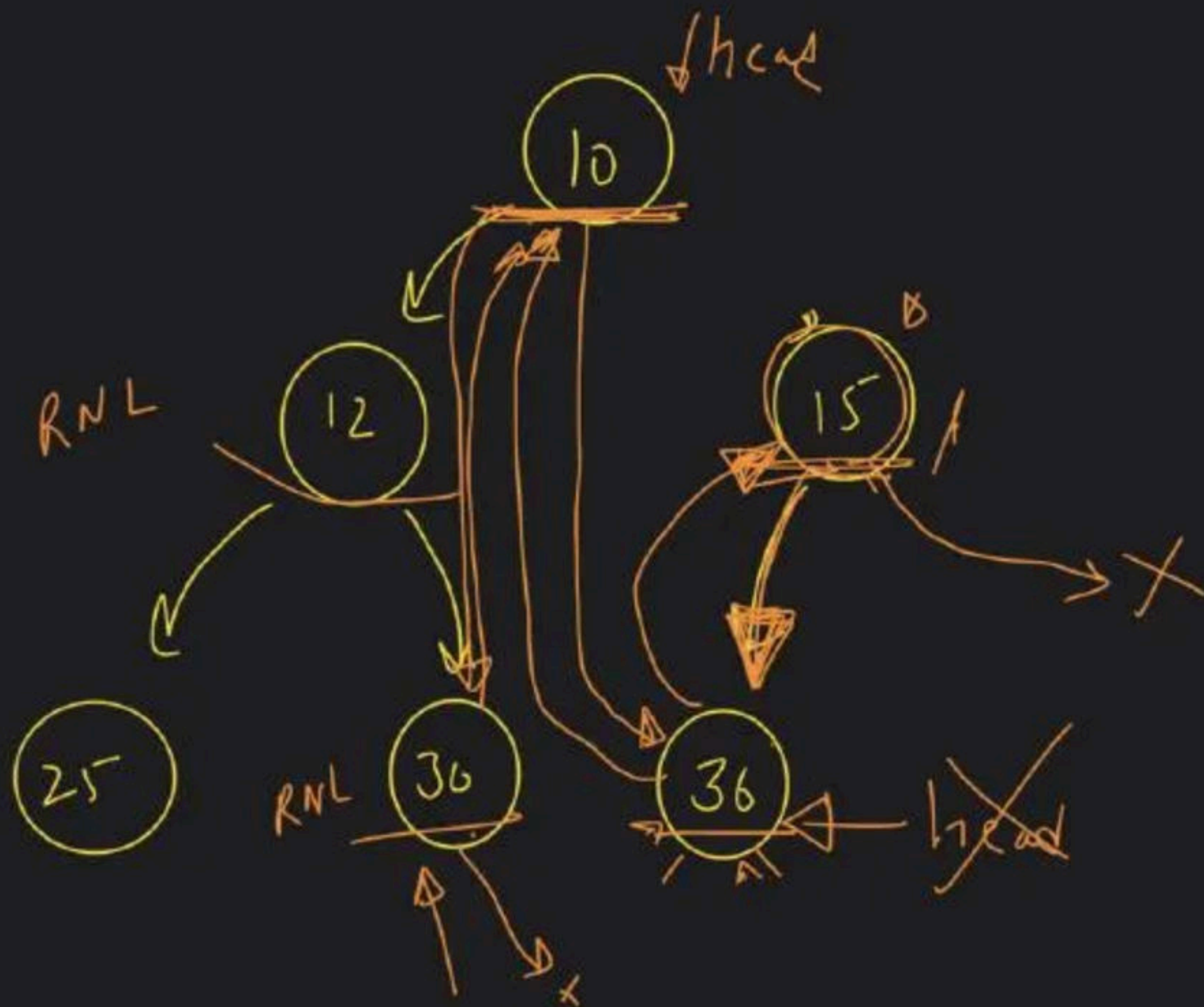
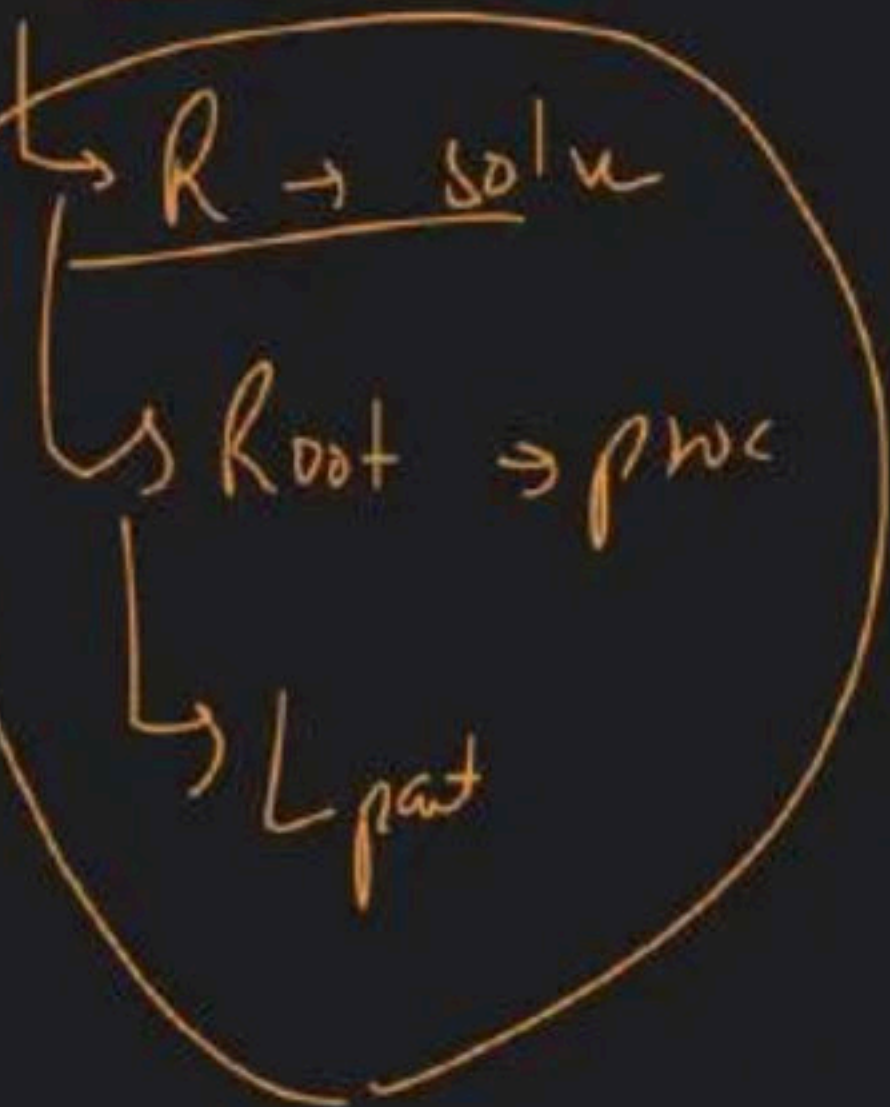


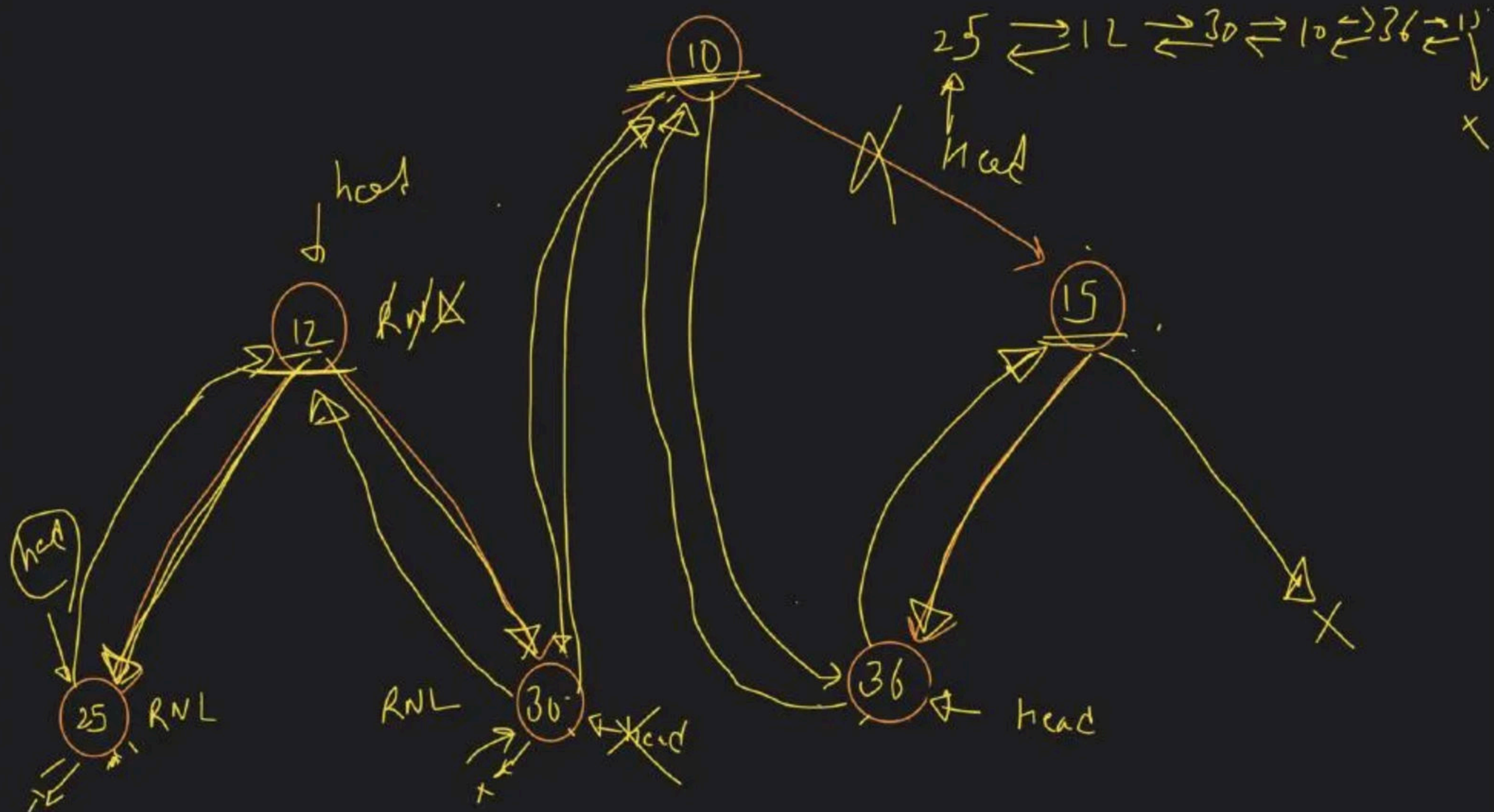
head = NULL



25 $\rightarrow$ 12 $\rightarrow$ 30 $\rightarrow$ 10 $\rightarrow$ 36 $\rightarrow$ 15 $\rightarrow$ X

Approach





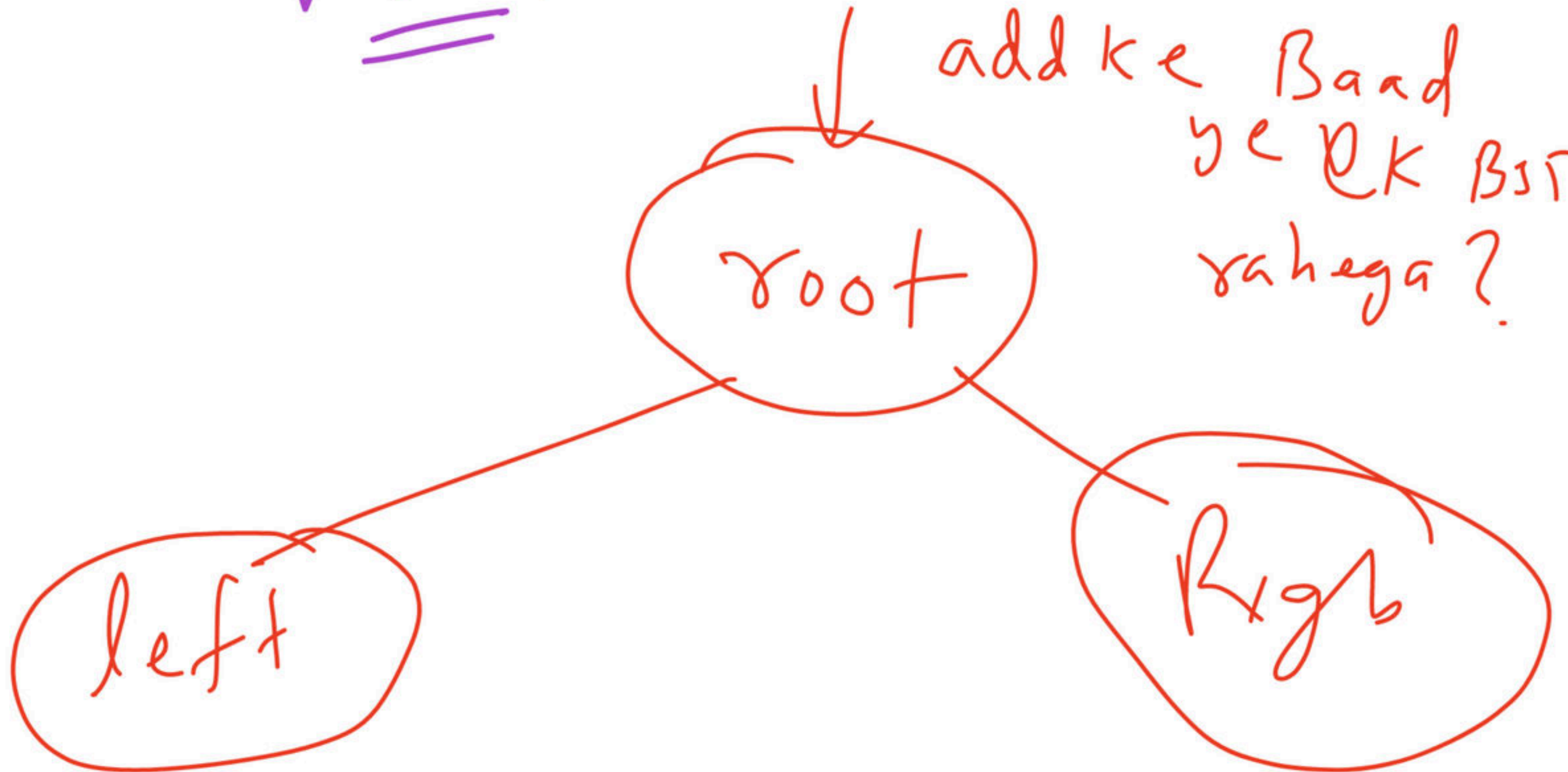
④

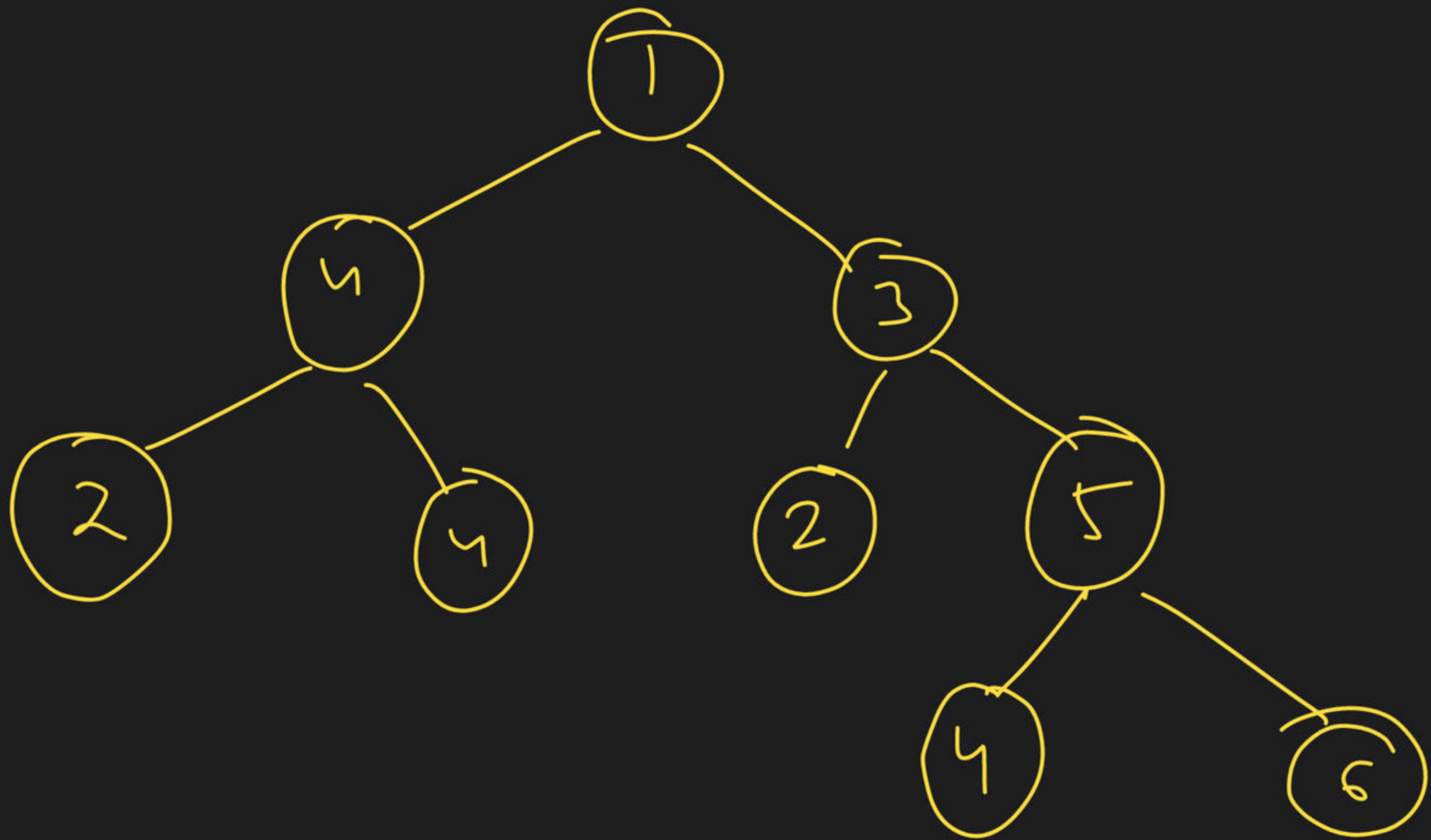
LC: 1373

max sum BST in
BT

V.V.VIMP

add ke Baad
ye ek BST
rahega?

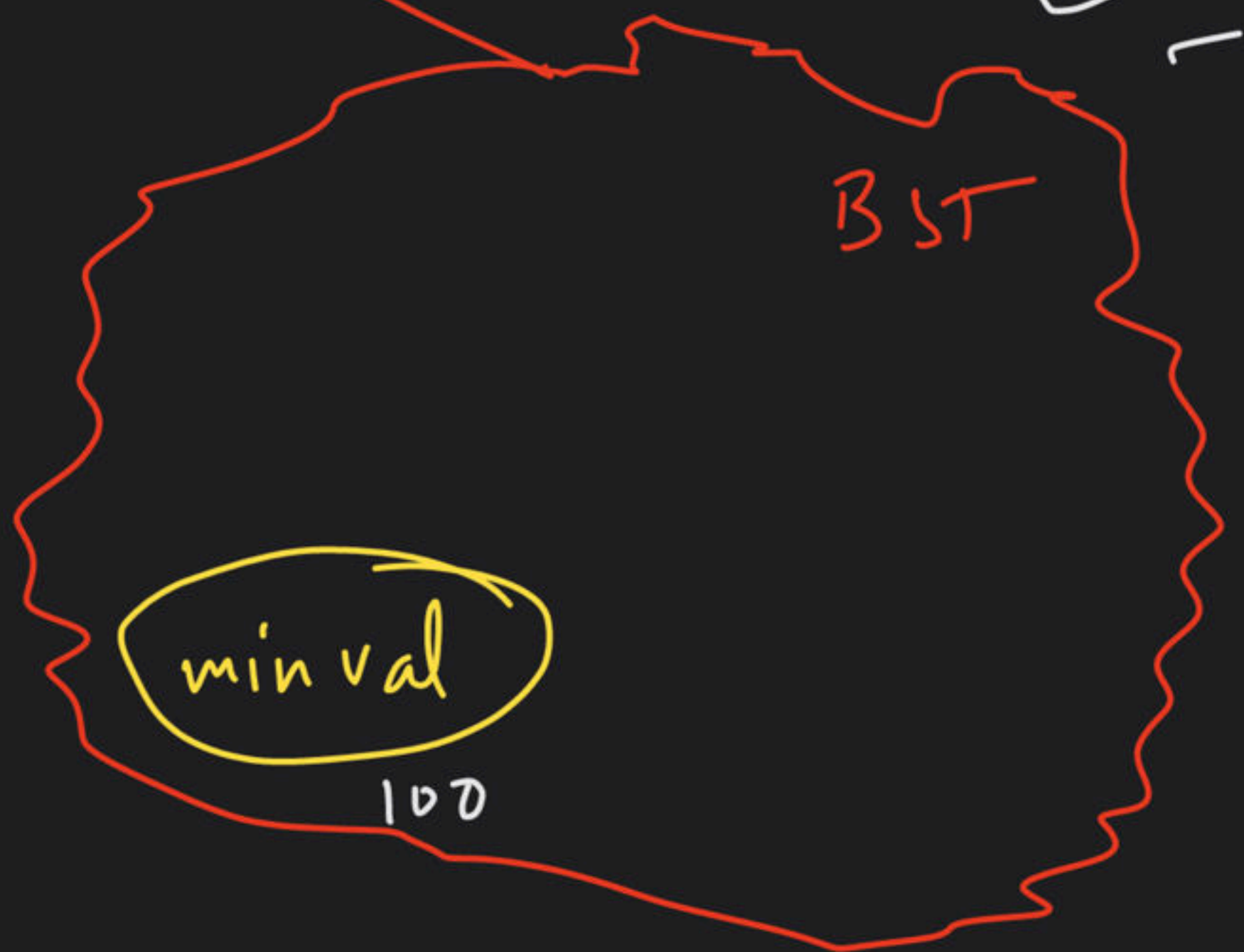
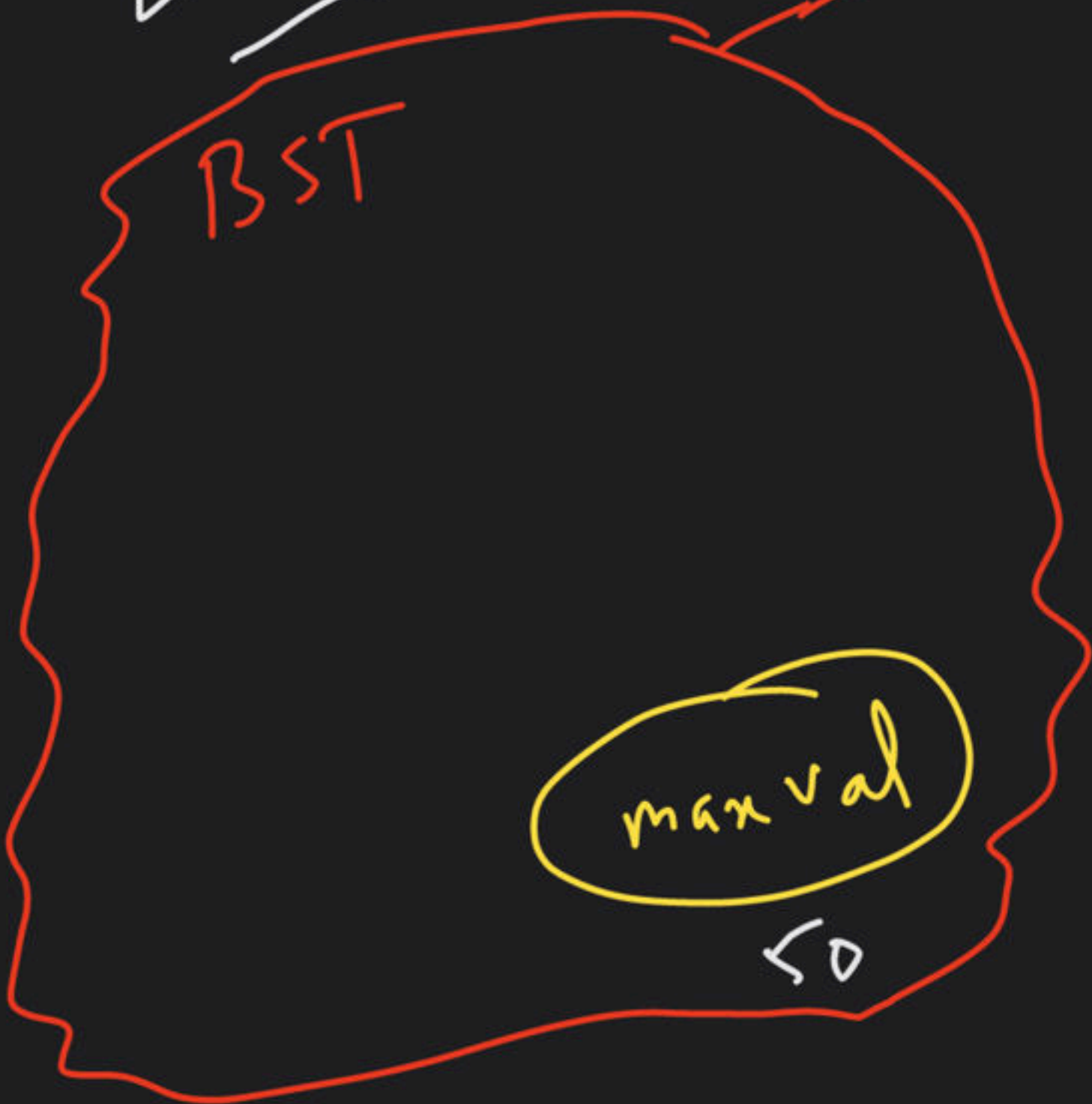




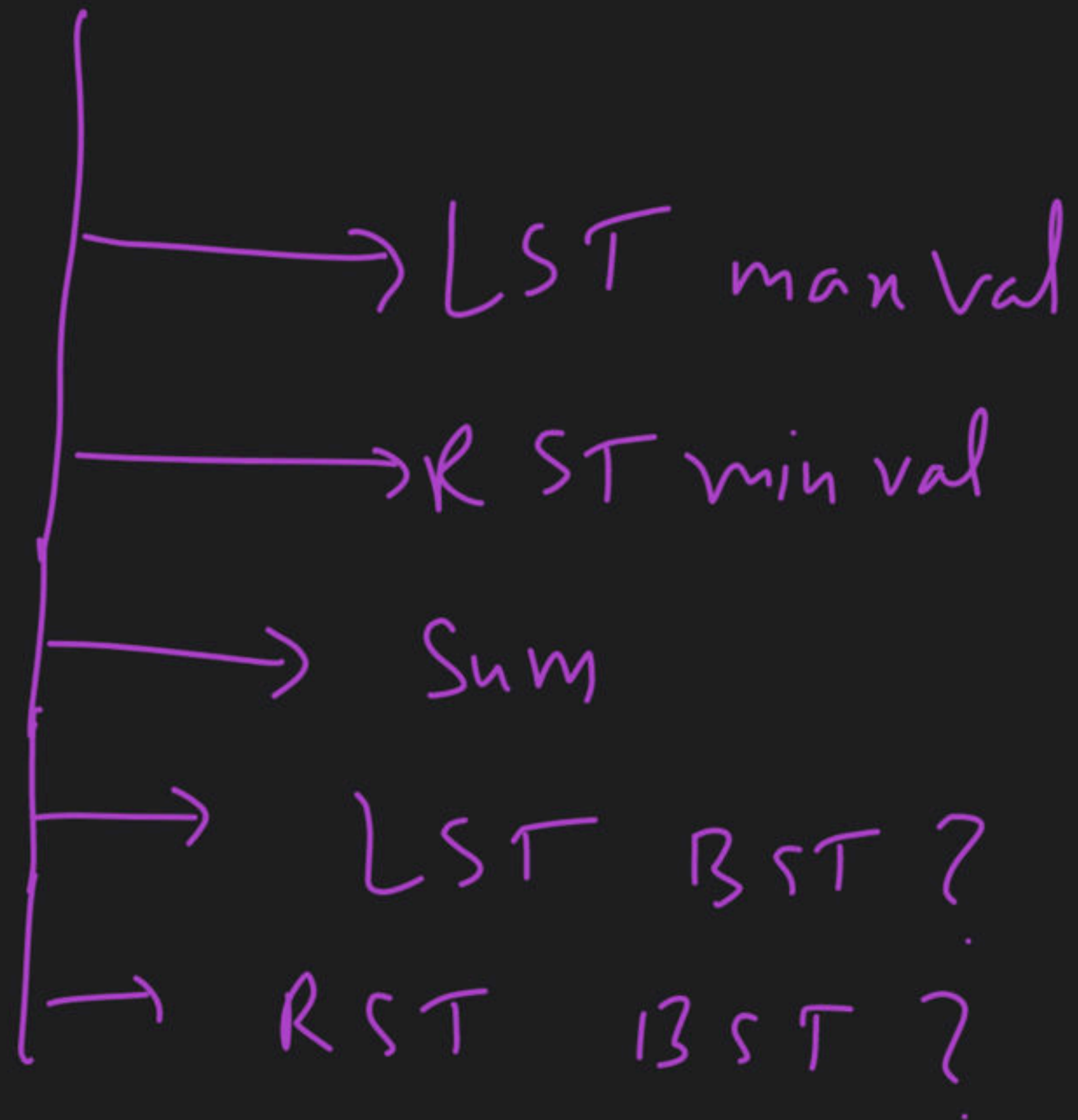
$X = 80$



$X > \text{LST max val}$
4 $X < \text{RST min val}$



① I want to keep track of LST
4 RST



→ leaf node

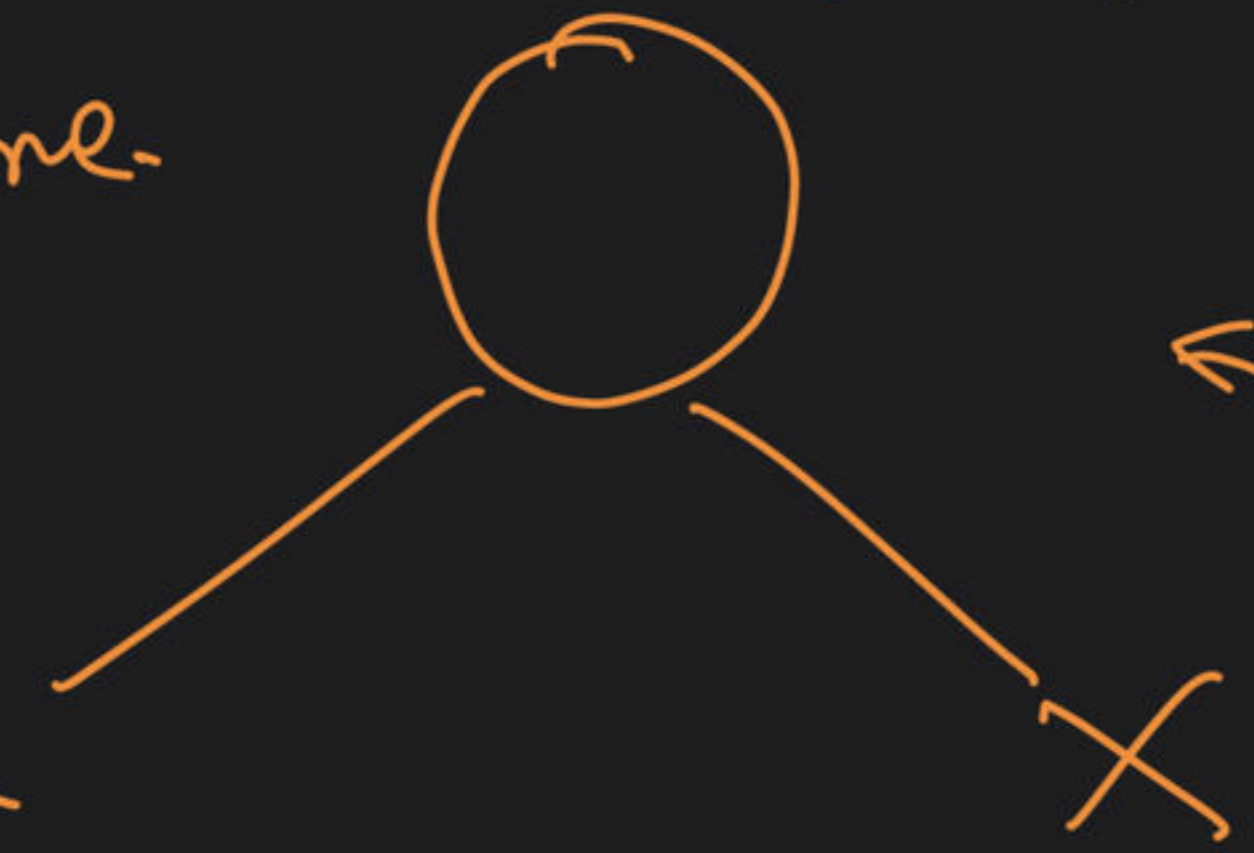
isBST tree.

isBST T_m

minVal $\Rightarrow$ INT Max

maxVal $\Rightarrow -\infty$

minVal $\Rightarrow \infty$
maxVal $\Rightarrow -\infty$



Null $\Rightarrow$ { isBST, sum, minVal, maxVal }

Leaf $\Rightarrow$ { T, 0, ∞ , $-\infty$ }

{ T, 2, 2, 2 }

Sum = 2

2

{ F, 0, 0, 0 }

{ T, 4, 4, 4 }

{ F, 0, 0, 0 }

Post Tra $\equiv$

{ T, 20, 2, 6 }

{ T, 15, 4, 6 }

{ T, 2, 2, 2 }

{ T, 6, 6, 6 }

{ T, 4, 4, 4 }

Max Sum = ~~6~~ ~~15~~ 20







